CONCRETE GRAND CHALLENGE

CIVITA: Computational Infrastructure for Validation, Integration, Testing, and Assessment of Concrete Computational Frameworks

DOCUMENTATION

Grand Challenge Working Group

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Part I

The CIVITA Network

Chapter 1

Introduction

Despite several decades of research, computational models for concrete have achieved only limited adoption in engineering practice. This limited translation is not due to a lack of scientific activity. On the contrary, the number of publications related to computational modeling of concrete has increased dramatically. Figure 1.1 shows that Scopus-indexed publications on computational modeling of concrete increased from only a few hundred papers per year in the late 1970s and early 1980s to about 1,500–2,000 papers per year around 2000, then grew rapidly to roughly 7,000–8,000 papers per year by 2012–2015, and accelerated further after 2020, reaching more than 20,000 publications per year in the most recent data.

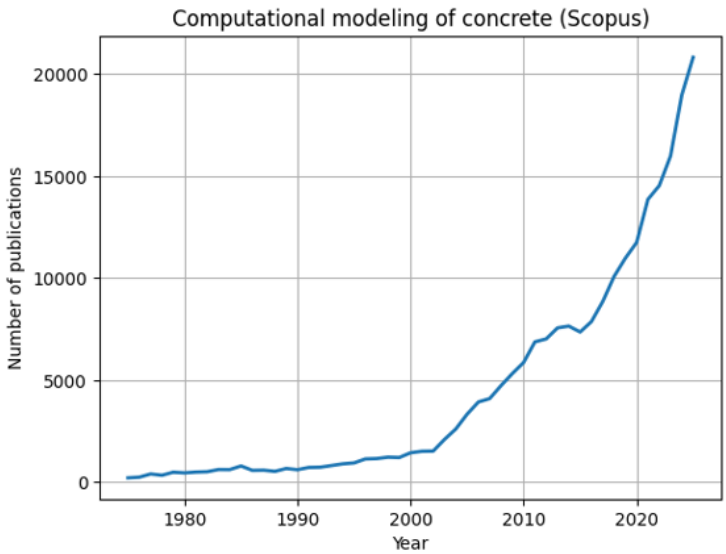


Figure 1.1: Growth of Scopus-indexed publications on computational modeling of concrete from the mid-1970s to the mid-2020s.

However, this growth has not been matched by a comparable increase in practical confidence, standardization, or routine use in engineering design. Significant intellectual and financial resources continue to be invested in the development of new models, refinements of existing formulations, and increasingly complex numerical implementations. Yet, many of these efforts remain incremental and are often not accompanied by rigorous validation, transparent comparison against experimental evidence, or systematic benchmarking against existing, competing approaches.

The central problem is the lack of a broadly accepted framework for assessing whether such models are reliable, predictive, and useful for practical applications. Without curated experimental databases, standardized validation protocols, and objective comparative assessments, the concrete modeling community risks continued growth in publication volume without corresponding progress in engineering adoption. This gap between research productivity and practical impact motivates the need for a coordinated effort focused on calibration, validation, benchmarking, and community-wide standards for concrete computational models.

This has motivated the launch of the *Grand Challenge on the Calibration, Validation, and Benchmarking of Concrete Computational Models*, a community-driven initiative aimed at improving the reliability, transparency, reproducibility, and predictive capability of computational modeling in concrete mechanics.

Chapter 2

CIVITA Architecture

CIVITA, the *Computational Infrastructure for Validation, Integration, Testing, and Assessment*, is an open scientific platform designed to support the systematic development, calibration, validation, and assessment of computational models for concrete and cementitious materials. It provides the digital infrastructure for the *Grand Challenge on the Calibration, Validation, and Benchmarking of Concrete Computational Models*.

The long-term vision of CIVITA is to establish a shared scientific environment in which experimental evidence, computational models, parameter sets, simulation results, and model-assessment procedures can be represented through structured and interoperable digital objects. Such an environment is intended to facilitate cumulative model development. Experimental datasets should be reusable across different modeling approaches; calibrated parameter sets should be transferable directly to validation exercises; validation results should provide quantitative information on model predictive capability; and standardized benchmarks should enable comparisons among model formulations, numerical implementations, and successive versions of computational frameworks.

CIVITA is organized around two complementary databases and three computational assessment environments:

1. The **Experimental Database** provides structured experimental data together with information describing materials, specimens, geometries, test configurations, loading histories, instrumentation, and measured quantities.

2. The **Computational Database** provides a structured repository of computational models, numerical implementations, model configurations, material parameter sets, and simulation results. It establishes the relationships among model formulations, software implementations, and their versions, allowing computational results to be associated unambiguously with the frameworks that produced them.

3. The **Parameter Identification** environment supports the calibration of model parameters against selected experimental observations. Calibration procedures and their results are documented to enable reproducibility and subsequent reuse.

4. The **Validation** environment supports prediction exercises in which previously calibrated models are assessed against independent experimental data. Its objective is to quantify predictive accuracy and establish the range of applicability of different computational models.

5. The **Benchmarking** environment provides standardized computational problems for comparing different versions, implementations, and formulations of computational models. Bench-

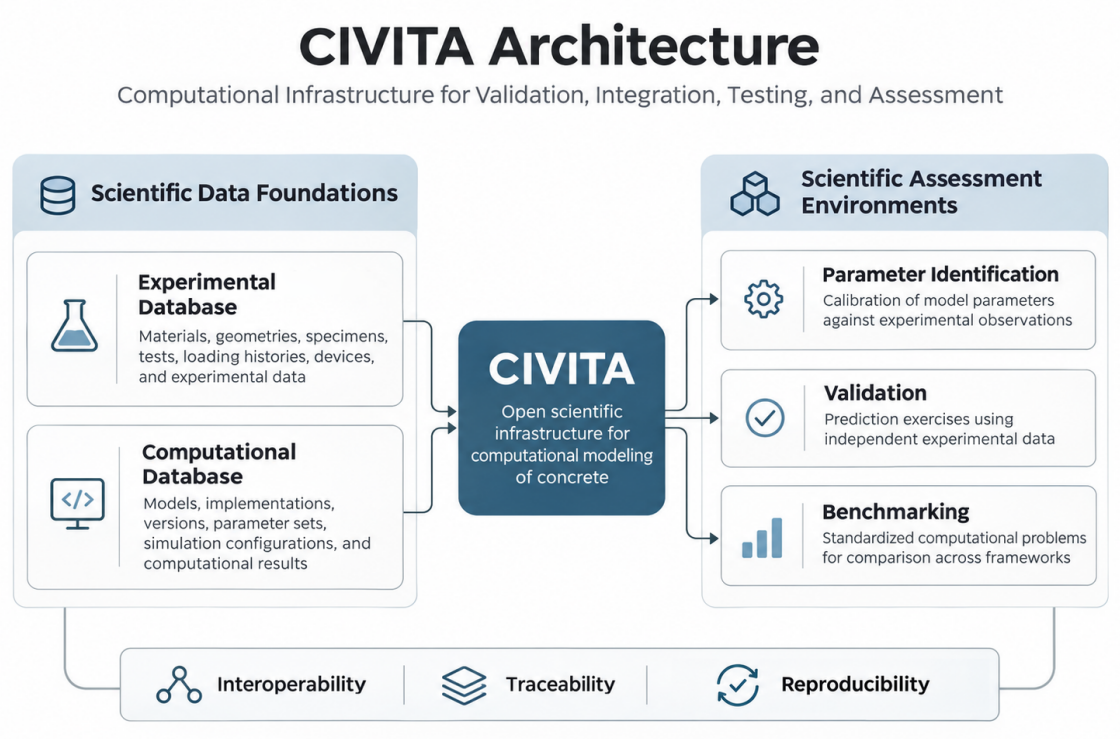


Figure 2.1: CIVITA architecture. Integration of experimental and computational databases with parameter identification, validation, and benchmarking environments to support reproducible assessment of concrete computational models.

mark specifications and submitted results are preserved to support reproducible comparisons and the progressive assessment of computational frameworks.

The two databases constitute the principal repositories of scientific evidence and computational knowledge within CIVITA. The Concrete Experimental Database preserves the physical evidence obtained from laboratory investigations, while the Computational Database organizes the models and numerical tools used to interpret and predict that evidence. Their interoperability provides the foundation for the three assessment environments, enabling experimental observations, computational configurations, and simulation results to be exchanged through consistent data structures.

The architecture of CIVITA is model-agnostic. It is intended to accommodate different constitutive theories, discrete and continuum formulations, multiscale approaches, and numerical implementations. A computational framework is identified by the combination of a mathematical model and its implementation in a specific software package. Distinguishing these two elements is essential for separating differences arising from model formulations from those associated with numerical implementations. CIVITA does not prescribe a preferred model. Its purpose is to provide the infrastructure required to evaluate models systematically and to document clearly the conditions under which their predictions are reliable.

A central principle of CIVITA is scientific traceability. Experimental datasets, computational models, software implementations, parameter configurations, calibration procedures, validation exercises, and benchmark results should remain identifiable, versioned, and reproducible. Open contributions are encouraged, while their provenance and review status should remain explicit. Over time, this structure is intended to create a persistent record of computational evidence from which

the capabilities and limitations of concrete models can be assessed quantitatively.

More broadly, CIVITA seeks to support a transition toward validation-centered computational engineering. As experimental and computational databases expand and scientific workflows become increasingly automated, including through artificial-intelligence agents, the value of computational models will depend increasingly on the quality of the evidence supporting their predictive capability. CIVITA is conceived as the infrastructure through which that evidence can be collected, connected, interrogated, and progressively strengthened by the research community.

Chapter 3

Shared Scientific Definitions

3.1 General conventions

3.2 Global Geometry Parametrization Library

The Geometry Parametrization Library defines the standardized parametric representations used to describe specimen geometries throughout the database. Its primary purpose is to convert heterogeneous geometric descriptions reported in the literature into consistent parameter sets. Each geometry type is represented through a predefined set of parameters, coordinate conventions, and dimensional descriptors stored in structured CSV or JSON-compatible formats. The parametrization framework enables consistent interpretation of specimen dimensions across datasets originating from different laboratories and experimental programs while minimizing redundancy and facilitating reuse of geometry definitions across multiple specimens and tests. The library is extensible, allowing additional geometry types and parameter definitions to be incorporated as new experimental configurations emerge during the data collection effort. The machine-readable definitions of the geometry parametrizations are additionally provided through the accompanying `geometry_library.json` schema file, which formalizes parameter names, units, coordinate conventions, required fields, and validation rules associated with each geometry type. Geometry parameter files may be provided either in CSV format or JSON format.

The geometry parametrization defines the specimen dimensions and associated coordinate system using a standardized right-handed Cartesian reference frame.

The current library includes:

- `cube`
- `cylinder`
- `dogbone`
- `double_notched_prism`
- `notched_prism`
- `ell_shape`
- `notched_cylinder`
- `prism`

3.2.1 `cube`

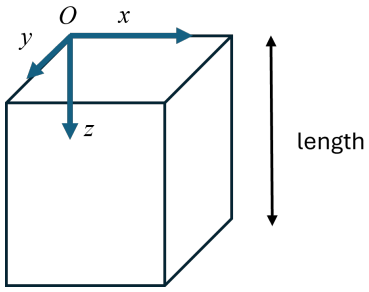


Figure 3.1: Cube parametrization

As illustrated in Fig. 3.1, the local coordinate system is defined by the axes x, y, and z, where the origin corresponds to the reference corner of the specimen. The x direction represents the longitudinal direction of the specimen, the y direction represents the transverse thickness direction, and the z direction represents the vertical depth direction.

The cube geometry is defined by the following parameter:

- `length`: side length of the cube.

3.2.2 `cylinder`

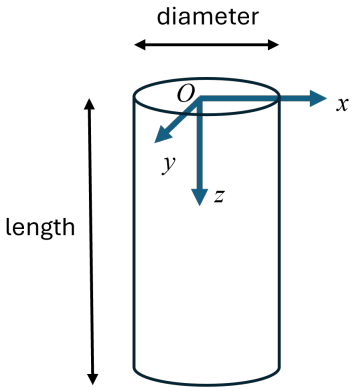


Figure 3.2: Cylinder parametrization

As illustrated in Fig. 3.2, the local coordinate system is defined by the axes x, y, and z, where the origin corresponds to the center of the top circular surface of the specimen. The z axis represents the longitudinal direction associated with the cylinder length, while the x and y axes define the transverse directions within the circular cross section.

The cylinder geometry is defined by the following parameters:

- `length`: longitudinal length of the cylinder measured along the z axis;
- `diameter`: diameter of the circular cross section.

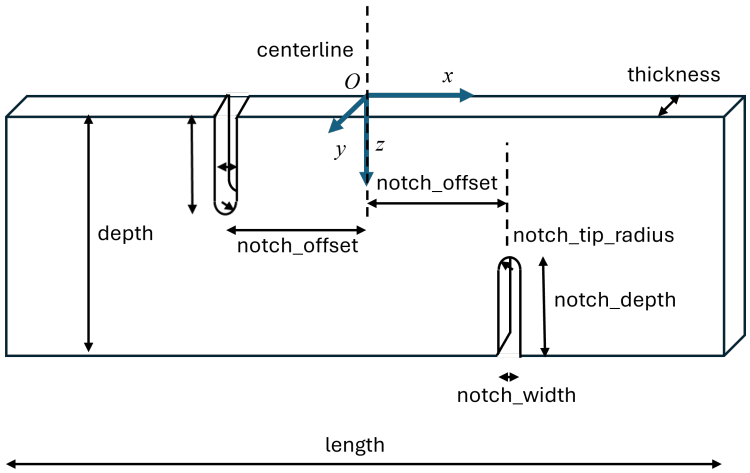


Figure 3.3: Doubly notched prism parametrization

3.2.3 `double_notched_prism`

As illustrated in Fig. 3.3, the local coordinate system is defined by the axes x, y, and z, where the origin is located on the centerline of the specimen. The x axis represents the longitudinal direction associated with the prism length, the y axis represents the transverse thickness direction, and the z axis represents the vertical depth direction. The two notches are positioned anti-symmetrically relative to the specimen centerline through the parameters `notch_offset`.

The double notched prism geometry is defined by the following parameters:

- `length`: prism length measured along the x axis;
- `thickness`: prism thickness measured along the y axis;
- `depth`: prism depth measured along the z axis;
- `notch_offset`: offset of the first notches relative to the specimen centerline;
- `notch_depth`: notch penetration depth measured along the z direction;
- `notch_width`: notch opening width;
- `notch_tip_radius`: radius at the notch tip.

3.2.4 `dogbone`

As illustrated in Fig. 3.4, the local coordinate system is defined by the axes x, y, and z, where the origin is located on the top side and on the longitudinal axis of symmetry of the specimen. The z axis represents the longitudinal direction associated with the dogbone length, the y axis represents the transverse thickness direction, and the x axis represents the horizontal width direction. The reduced central section of the specimen is defined through the corresponding geometric parameters specified below.

- `length`: total specimen length measured along the z axis;
- `width`: maximum specimen width measured along the x axis;

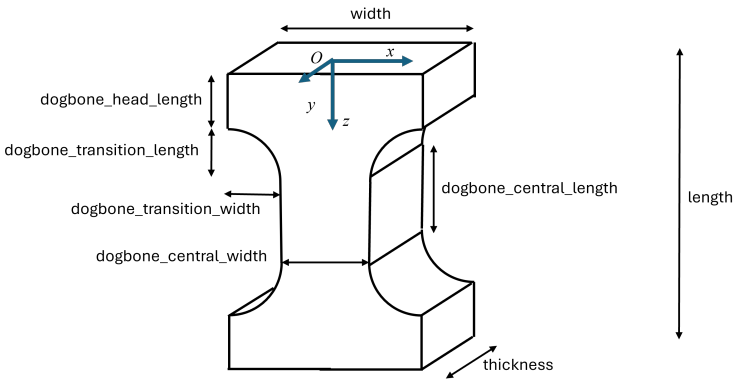


Figure 3.4: Dogbone parametrization

- `thickness`: specimen thickness measured along the y axis;
- `dogbone_central_length`: length of the reduced central region measured along the z axis;
- `dogbone_central_width`: width of the reduced central region measured along the x axis;
- `dogbone_head_length`: length of each enlarged head region measured along the z axis;
- `dogbone_transition_length`: length of each transition region between the head and central regions measured along the z axis;
- `dogbone_transition_width`: width variation defining the transition from the head width to the central width measured along the x axis.

3.2.5 `notched_prism`

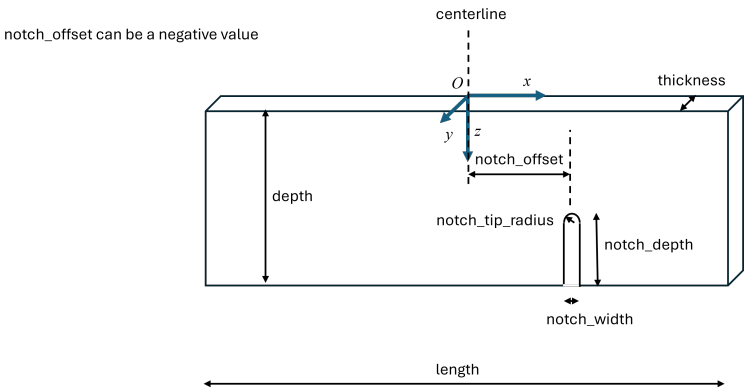


Figure 3.5: Notched prism parametrization

As illustrated in Fig. 3.5, the local coordinate system is defined by the axes x, y, and z, where the origin corresponds to the reference corner of the specimen. The x axis represents the longitudinal direction associated with the prism length, the y axis represents the transverse thickness direction, and the z axis represents the vertical depth direction. The notch is positioned relative to the specimen centerline through the parameter `notch_offset`. `notch_offset` can be either positive or negative.

The notched prism geometry is defined by the following parameters:

- `length`: prism length measured along the x axis;
- `thickness`: prism thickness measured along the y axis;
- `depth`: prism depth measured along the z axis;
- `notch_offset`: offset of the notch relative to the specimen centerline;
- `notch_depth`: notch penetration depth measured along the z direction;
- `notch_width`: notch opening width;
- `notch_tip_radius`: radius at the notch tip.

3.2.6 `ell_shape`

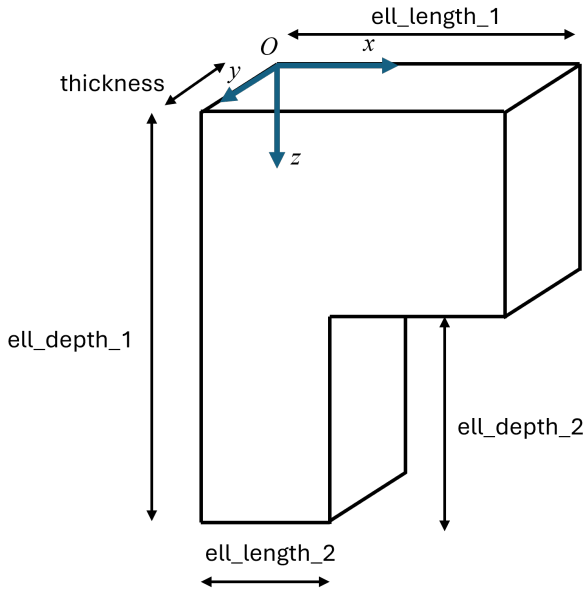


Figure 3.6: Ell shape parametrization

As illustrated in Fig. 3.6, the local coordinate system is defined by the axes x, y, and z, where the origin corresponds to the reference corner of the specimen. The x axis represents the longitudinal direction of the specimen, the y axis represents the transverse thickness direction, and the z axis represents the vertical depth direction. The L-shaped geometry is defined through the corresponding geometric parameters specified in the geometry parametrization library.

The ell-shape geometry is defined by the following parameters:

As illustrated in Fig. 3.6, the local coordinate system is defined by the axes x, y, and z, where the origin corresponds to the reference corner of the specimen. The x axis represents the longitudinal direction of the specimen, the y axis represents the transverse thickness direction, and the z axis represents the vertical depth direction. The L-shaped geometry is defined through the corresponding geometric parameters specified below.

- `ell_depth_1`: depth of the vertical branch measured along the z axis;

- `ell_depth_2`: depth of the horizontal branch measured along the z axis;
- `ell_length_1`: length of the horizontal branch measured along the x axis;
- `ell_length_2`: length of the vertical branch measured along the x axis;
- `thickness`: specimen thickness measured along the y axis.

3.2.7 `notched_cylinder`

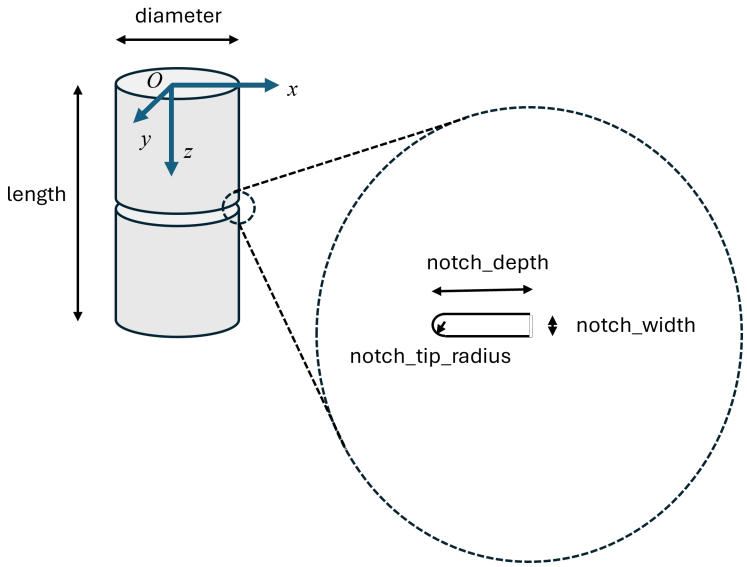


Figure 3.7: Notched cylinder parametrization

As illustrated in Fig. 3.7, the local coordinate system is defined by the axes x, y, and z, where the origin corresponds to the center of the top circular surface of the specimen. The z axis represents the longitudinal direction associated with the cylinder length, while the x and y axes define the transverse directions within the circular cross section. The circumferential notch is positioned along the cylinder axis at mid-length.

The notched cylinder geometry is defined by the following parameters:

- `length`: longitudinal length of the cylinder measured along the z axis;
- `diameter`: diameter of the circular cross section;
- `notch_depth`: radial notch depth;
- `notch_width`: axial width of the circumferential notch;
- `notch_tip_radius`: radius at the notch tip.

3.2.8 `prism`

As illustrated in Fig. 3.8, the local coordinate system is defined by the axes x, y, and z, where the origin corresponds to the reference corner of the specimen. The x axis represents the longitudinal

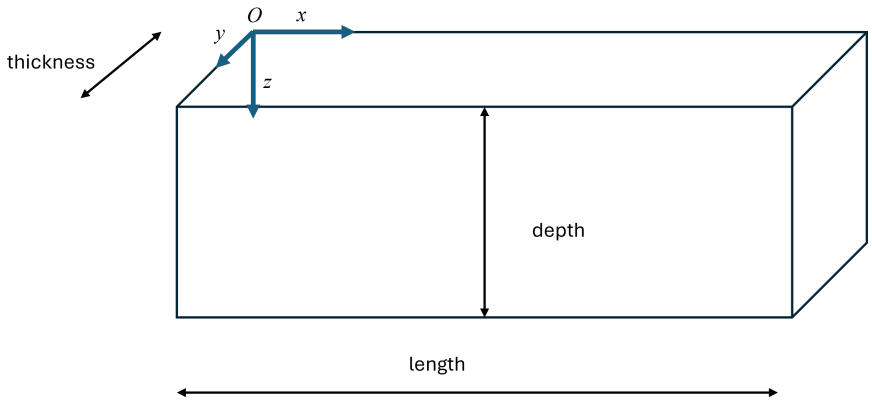


Figure 3.8: Prism parametrization

direction associated with the prism length, the y axis represents the transverse thickness direction, and the z axis represents the vertical depth direction.

The prism geometry is defined by the following parameters:

- `length`: prism length measured along the x axis;
- `thickness`: prism thickness measured along the y axis;
- `depth`: prism depth measured along the z axis.

Part II

Experimental Database

Chapter 4

Introduction

4.1 Vision

This document describes the structure, terminology, workflow, and file standards for submitting datasets to the Concrete Grand Challenge Experimental Database. The database structure is primarily designed to support calibration, validation, and benchmarking of concrete computational models through a curated, traceable, and machine-readable collection of experimental data.

The development of the experimental database will proceed in multiple phases, beginning with the current data collection and standardization effort. At this stage, the primary objective is to assemble a large, well-structured, and community-driven repository of experimental data relevant to concrete materials, structural components, and associated physical phenomena. The data are currently being curated through standardized spreadsheet templates that organize information into categories such as literature sources, keywords, material batches, specimen definitions, specimen geometries, reinforcement configurations, test setups, loading histories, measurement devices, printing variables, and experimental observations. The framework is designed to accommodate a broad range of data types, including scalar quantities, curves, tables, images, videos, and multidimensional fields. The current database structure separates reusable specimen-level information from test-level information to reduce redundancy and improve consistency across datasets. The specimen geometry, reinforcement layout, test setups, and devices & supports schemes currently adopted are based on standardized parametric representations defined in dedicated parametrization libraries.

Furthermore, a major focus of this initial phase is the creation of controlled vocabularies, standardized metadata definitions, and machine-readable schema layers. This includes the use of unique identifiers, hierarchical relationships between entities, predefined taxonomies for mechanisms and test types, standardized parametrization schemes, and consistent naming conventions for files and folders. The goal is to ensure that the collected information is machine-readable, interoperable, and suitable for large-scale data analytics, model calibration, validation, benchmarking, and future AI-assisted workflows. During this phase, data contributors can continue using familiar spreadsheet-based workflows while progressively converging toward a unified and extensible data structure.

An additional component of the evolving infrastructure is the development of machine-readable schema and vocabulary definition layers based on structured formats such as JSON. These files will provide formal definitions of controlled vocabularies, metadata schemas, parameter descriptions, validation rules, unit conventions, parametrization libraries, and hierarchical relationships between database entities. The JSON layer will support automated validation, interoperability with external software platforms, API development, AI-assisted curation workflows, semantic search, ontology development, and future agent-based systems for automated data extraction and interpretation. The

use of portable machine-readable schema definitions will also facilitate version control, extensibility, and synchronization between spreadsheet templates, database structures, parametrization libraries, and software tools interacting with the database.

Once the initial data collection effort reaches sufficient maturity, the database will transition to a more robust and scalable architecture based on a PostgreSQL relational database coupled with HDF5 storage for large numerical datasets. In this architecture, PostgreSQL will manage all metadata, indexing, relationships, provenance information, controlled vocabularies, parametrization libraries, schema definitions, search capabilities, and user queries, while HDF5 files will store computationally intensive data such as time histories, multidimensional fields, large tables, images, videos, and other high-volume experimental outputs. JSON-based schema and vocabulary layers will complement this architecture by providing structured machine-readable definitions and validation specifications interfacing with both the relational database and scientific data storage components. This hybrid approach combines the flexibility and query efficiency of relational databases with the performance and scalability of HDF5 for scientific data storage while maintaining extensible, interoperable, and machine-readable metadata standards.

4.2 Concrete Grand Challenge Working Group

The following table lists the current members of the Concrete Grand Challenge Working Group together with their institutional affiliations and contact information. This initial version of the database structure and collection strategy is the result of the concerted effort of this group. Particular credit goes to **M.Z. Naser** (Clemson University, `mznaser@clemson.edu`) for the development of the AI-assisted curation workflow.

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Chapter 5

Database Philosophy

The database is guided by a set of core principles intended to ensure scientific rigor, long-term usability, interoperability, and compatibility with future computational workflows. In particular, the database architecture is designed to align as closely as possible with the FAIR data principles, namely that data should be **Findable**, **Accessible**, **Interoperable**, and **Reusable**. These principles are implemented through standardized identifiers, controlled vocabularies, machine-readable metadata, explicit provenance tracking, and extensible parametrization libraries.

The database follows the following main principles.

1. **Traceability and Provenance**. Every data item must be traceable to the original literature source. Provenance information such as figure numbers, table references, extraction dates, and curator information should also be preserved whenever possible. This principle is essential to guarantee reproducibility, scientific transparency, and long-term reusability of the data.

2. **Findability**. All entities within the database are identified through standardized unique identifiers and organized according to a consistent hierarchical structure. Controlled vocabularies, standardized naming conventions, keywords, and machine-readable metadata facilitate indexing, semantic search, filtering, ontology integration, and automated discovery of relevant datasets across different experimental programs and laboratories.

3. **Accessibility**. The database structure is designed to support long-term accessibility of both metadata and associated experimental files. Metadata are stored in structured spreadsheet and future relational-database formats, while large data objects such as curves, tables, images, videos, and multidimensional datasets are stored as external files referenced through persistent identifiers and standardized folder structures. The framework is intended to support future API-based access, automated querying, and cloud-based data services.

4. **Interoperability and Machine Readability**. Data are represented using standardized parametrizations, controlled vocabularies, explicit units, machine-readable schema layers, and consistent file formats. The use of structured JSON schema and vocabulary definition files facilitates interoperability with external software platforms, ontology frameworks, computational workflows, digital twin infrastructures, and AI-assisted systems. This principle also enables automated parsing, validation, semantic indexing, and machine learning applications.

5. **Reusability**. The database is organized to maximize long-term scientific reuse of the data across different applications, including computational model calibration, validation, benchmarking, uncertainty quantification, and machine learning. Standardized parametrization

schemes promote consistency and facilitate direct reuse of experimental information in computational workflows.

6. **Separation of Metadata and Data Files**. The spreadsheet and future relational database layers primarily store metadata, structured descriptors, identifiers, provenance information, and references to external files. Large experimental datasets, such as curves, tables, images, videos, multidimensional fields, and voxel data, are stored separately in dedicated files using standardized formats such as CSV, JSON, HDF5, images, or video formats. This separation improves scalability, maintainability, and computational efficiency.

7. **Extensibility and Community-Driven Evolution**. The database is designed to evolve over time. Contributors are encouraged to propose additional geometry types, test parametrizations, reinforcement configurations, observed mechanisms, device types, material descriptors, quantity types, ontology mappings, and schema extensions whenever the current libraries and vocabularies are insufficient. The parametrization libraries and machine-readable schema layers are therefore treated as living components of the database infrastructure.

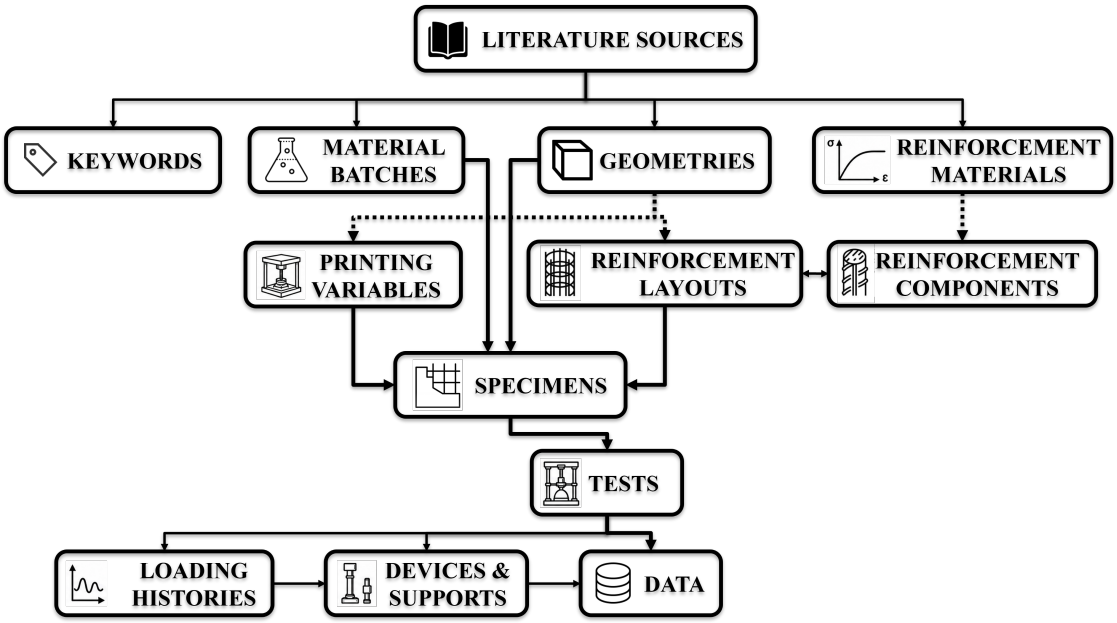


Figure 5.1: Hierarchical organization of the database showing the relationships between literature sources, material batches, geometries, reinforcement materials, reinforcement components, reinforcement layouts, specimens, tests, devices and supports, loading histories, and experimental data entries.

5.1 Database Hierarchy

The database is organized according to a hierarchical structure designed to preserve the logical relationships between literature sources, material systems, specimen definitions, experimental tests, and associated datasets (Fig. 5.1). At the highest level, each literature source represents a publication, report, thesis, dataset, or experimental campaign from which the data originate. Literature sources are linked to keywords that facilitate indexing, search, and categorization, as well as to reusable

definitions of material batches, geometries, reinforcement systems, printing variables, specimens, tests, and associated datasets.

Material batches describe the specific concrete mixtures or material systems investigated within a source. Geometries define reusable specimen geometrical configurations through standardized parametrizations provided in the geometry parametrization library. Printing variables provide reusable definitions relevant to additively manufactured specimens.

The reinforcement hierarchy is organized through three distinct but interconnected entities: reinforcement materials, reinforcement components, and reinforcement layouts. Reinforcement materials define reusable material properties associated with reinforcing systems. Reinforcement components represent individual reinforcement entities, such as longitudinal bar layers, stirrup sets, tie systems, or externally bonded reinforcement elements, and combine reinforcement geometry with the associated reinforcement material definition. Reinforcement layouts and components are associated with a specific geometry. Multiple reinforcement components are then assembled into reusable reinforcement layouts representing complete reinforcement configurations for specimens.

A specimen represents the combination of a material batch, geometry, reinforcement layout (if relevant), and optional printing variables (if relevant) associated with a physical specimen or specimen family.

Specimens are connected to individual tests that contain detailed information about the experimental procedures, environmental conditions, loading configurations, and observed physical mechanisms. Each test serves as the central entity to which additional components are attached, including devices and supports, loading histories, and the collected experimental data. Devices and supports may represent actuators, sensors, fixtures, supports, instrumentation systems, or measurement devices associated with the test setup. Loading histories define loading protocols or time histories associated with a test and, when relevant, with a specific device.

Experimental observations are organized through the data entity, which may include scalar quantities, curves, tables, images, videos, and multidimensional datasets together with their associated provenance and metadata. Data entries may optionally be linked to specific devices and supports when the measurements originate from a particular instrumentation component.

5.2 Parametrization Schemes

A major challenge in building a large-scale experimental database for concrete materials and structures is the enormous variability in specimen configurations, reinforcement layouts, boundary conditions, and testing procedures reported in the literature. Experimental programs are often described using highly heterogeneous terminology, inconsistent geometric definitions, incomplete drawings, or source-specific conventions that are difficult to interpret computationally. While this level of flexibility may be acceptable for traditional human interpretation of journal articles and reports, it creates severe limitations for machine readability and automated data extraction.

To address these challenges, the database adopts **standardized parametrization schemes** for

1. specimen geometries
2. test setups
3. reinforcement layouts
4. devices, and supports configurations

These parametrization schemes define a limited but extensible set of canonical representations capable of describing the vast majority of experimental configurations encountered in concrete and reinforced concrete mechanics research. The parametrizations are organized through dedicated geometry, test, reinforcement, devices, and supports libraries, described later in this documentation, that define standardized parameter names, coordinate conventions, and required descriptors.

The adoption of parametrized representations provides several important advantages.

1. **Machine readability and interoperability.** Standardized parametrizations transform heterogeneous experimental descriptions into structured machine-readable data. This enables automated parsing, validation, indexing, and integration across datasets originating from different laboratories, countries, and historical periods.

2. **Consistency across datasets.** Canonical parameter names and coordinate conventions reduce ambiguity and minimize inconsistent interpretations of geometry dimensions, reinforcement locations, support conditions, and loading configurations.

3. **Efficient storage and reduced redundancy.** Parametric descriptions require only a compact set of defining variables instead of large unstructured textual descriptions or multiple redundant drawings. The same parametrized geometry can also be reused across multiple tests.

4. **Automated generation of computational models.** Parametrized representations facilitate automatic generation of computational models . This capability is particularly important for future automated calibration, validation, and benchmarking workflows.

5. **Support for machine learning applications.** Machine learning models require consistent numerical descriptors rather than free-form textual descriptions. Parametrized geometries and reinforcement schemes provide structured low-dimensional feature representations that can be directly used for classification, clustering, regression, surrogate modeling, and uncertainty quantification.

6. **Scalability and future extensibility.** The parametrization libraries are designed to evolve over time. New geometry types, reinforcement configurations, and test setups can be added while preserving compatibility with existing datasets and workflows.

7. **Improved traceability and reproducibility.** Standardized parametrizations make it easier to reproduce experiments computationally and verify the consistency of reported dimensions, reinforcement layouts, and loading configurations.

The parametrization schemes adopted in the current version of the database are designed to balance simplicity, extensibility, and practical usability. The objective is not to reproduce every possible geometric detail found in experimental programs, but rather to capture the dominant features required for computational modeling, data analytics, and systematic comparison across datasets. When experimental configurations cannot be adequately represented by the current libraries, contributors are encouraged to propose extensions and additional parametrization types as part of the ongoing evolution of the database framework.

The absence of an existing parametrization scheme for a particular geometry, reinforcement configuration, test setup, or device/support configuration is not a reason for exclusion from the database. In such cases, contributors should provide the most complete description possible using available metadata, drawings, images, parameter files, and notes, while also proposing potential extensions to the corresponding parametrization libraries.

5.3 Manual Data Submission

Each submission must contain a single dataset only, which will be automatically appended to the overall database after validation and processing. Contributors should submit one dataset at a time to simplify quality control, provenance tracking, and integration into the centralized repository. All identifiers must follow the prescribed naming convention and begin with the numerical sequence `0001` for each new submission category, for example `SRC0001`, `BAT0001`, and `TST0001`. Contributors are expected to populate all relevant cells in the spreadsheet template as completely and accurately as possible.

The submitted dataset must follow the required folder hierarchy and include all relevant supporting files associated with the experimental program. The top-level directory must be named using the source identifier and should contain the completed spreadsheet template together with an optional Python script or Jupyter Notebook capable of plotting and visualizing all experimental curves included in the dataset. The general folder structure is illustrated below:

```
SRC0001/
    SRC0001.xlsx
    SRC0001.ipynb
    BAT0001/
        TST0001/
        TST0002/
        .../
    BAT0002/
        TST0003/
        .../
```

Within each source directory, the spreadsheet stores the metadata and structured information associated with the experimental campaign, while the batch and test subdirectories contain all corresponding files such as CSV tables, loading histories, images, videos, geometry descriptions, measurement data, and other relevant artifacts. Each material batch is represented by a dedicated folder, and each experimental test associated with that batch is stored in its own test folder. Test numbering must remain progressive across all batches belonging to the same source.

Only data and files that can be legally redistributed should be included in the submission. Contributors must not upload copyrighted articles, scanned book chapters, or any other copyrighted material for which redistribution rights are unavailable. The submission should contain only the curated data, associated metadata, and supporting files necessary for interpretation and reuse of the experimental results.

After preparing the dataset and verifying the folder structure, contributors should compress the entire submission into a single ZIP archive using the following naming convention:

```
lastname_year_month_day_n.zip
```

For example:

```
cusatis_2026_4-18_1.zip
cusatis_2026_4-18_2.zip
...
```

The final ZIP archive should then be uploaded through the designated submission portal (*under development*). Additional information regarding the upload procedure and repository access will be provided soon.

During ingestion into the centralized database, all local identifiers will be automatically mapped to globally unique database identifiers.

5.4 AI-Assisted Data Submission

under development

5.5 Use of AI Tools in Documentation and Schema Development

This documentation was prepared with the assistance of artificial intelligence (AI) tools used to support drafting, editing, formatting, organizational tasks, controlled vocabulary definition suggestions, and generation of machine-readable JSON schema and library files. AI-assisted tools also contributed to consistency checking. All scientific, technical, and methodological content, including database architecture, parametrization schemes, controlled vocabularies, validation logic, and JSON definitions, was reviewed, validated, and curated by members of the Grand Challenge Working Group.

Given the scale, evolving nature, and technical complexity of the effort, the documentation and associated schema definitions may still contain inconsistencies, ambiguities, omissions, or unintended errors. Contributors and users are encouraged to report issues, suggest improvements, and propose extensions as part of the ongoing community-driven development of the database framework.

5.6 Naming Conventions

The following table summarizes the standardized identifier conventions adopted throughout the database to uniquely define the primary entities, metadata components, parametrization objects, and experimental data entries used in the submission workflow and database architecture.

Entity	**ID Format**	**Meaning**
Source	SRC0001	One publication, report, thesis, dataset, or experimental campaign source
Material batches	BAT0001	One material batch, mixture, or material system definition
Geometries	GEO0001	One specimen geometry definition as defined in the geometry parametrization library
Printing variables	PRI0001	One set of printing variables associated with an additively manufactured specimen

Reinforcement materials	RMT0001	One reusable set of material properties associated with reinforcement systems
Reinforcement components	REF0001	One reinforcement component, such as a reinforcement layer, stirrup set, or external reinforcement component, defined through the reinforcement parametrization library and associated with a reinforcement material
Reinforcement layouts	RFL0001	One reinforcement layout combining multiple reinforcement components
Specimens	SPC0001	One specimen definition combining material batch, geometry, reinforcement layout (if relevant), and printing variables (if relevant)
Tests	TST0001	One experimental test or environmental exposure condition
Devices & Supports	DEV0001	One support, actuator, sensor, instrumentation component, or measurement device
Loading histories	LDG0001	One loading protocol or loading history associated with a test
Data	DAT0001	One scalar quantity, curve, table, image, video, or multidimensional dataset entry

File names should be descriptive, machine readable, and should not contain spaces or special characters other than underscores, hyphens, and periods. Recommended separators are underscores. File names should also include the relevant entity identifier whenever possible to simplify traceability and automated parsing.

Examples include:

```
TST0001_load_displacement.csv
TST0001_temperature_history.csv
TST0001_test_setup.png
GEO0001_geometry_parameters.csv
REF0001_stress_strain_curve.csv
BAT0001_mixing_protocol.txt
DAT0001_crack_pattern_image.png
```

Contributors should avoid ambiguous file names such as:

```
data.csv
image1.png
testfinal.csv
```

Whenever possible, file names should reflect the physical meaning of the data, the associated entity, and the quantity being reported.

5.7 Coordinate Systems

All spatial information is defined using a right-handed Cartesian coordinate system with coordinates x, y, and z, where the origin and axis orientations are specified according to the reference configurations provided in the specimen geometry library.

5.8 Units Policy

All numerical values must include associated units, either directly in the spreadsheet or in the second row of the corresponding CSV file. Certain spreadsheet entries require predefined units, which are explicitly indicated in the column header. For example, the cell with header `cement_content_kg_m3` in the batch description sheet requires the cement content to be reported in units of kg/m^3. Contributors should strictly follow the prescribed units whenever they are specified in the column names to ensure consistency and compatibility across the database.

Quantity	**Preferred Unit**	**Notes**
Length	mm	Geometry, device locations, crack openings, displacements
Force	N or kN	Use consistently within each file
Stress	MPa	Preferred for concrete and reinforcement strength
Strain	dimensionless	Use decimal strain, not percent, unless explicitly stated
Density	kg/m^3	Material constituents
Temperature	°C	Environmental and thermal histories
Relative humidity	%	Environmental histories
Time	s, min, h, or day	State explicitly
Mass content	kg/m^3	Mix design quantities
Volume fraction	dimensionless or %	State explicitly
...	...	...

5.9 File Standards

5.9.1 Curve CSV Formatting

Curve CSV files must include two header rows followed by numerical values. The first row contains variable names. The second row contains units.

```
displacement,load
mm,kN
0.000,0.000
0.010,1.250
0.020,2.410
```

For multiple curves, create separate files.

5.9.2 Table CSV Formatting

Table CSV files must include two header rows. The first row contains column names. The second row contains units. Leave the unit cell blank for unitless or string data.

```
specimen,compressive_strength,elastic_modulus
,MPa,MPa
```

```
A1,42.1,31000
A2,41.7,30500
```

5.9.3 Specimen Geometry Parametsrs CSV Formatting

Geometry CSV files must contain three rows:

1. parameter names;
2. parameter units;
3. parameter values.

Example for a notched beam:

```
length,depth,thickness,notch_depth,notch_width,notch_tip_radius
mm,mm,mm,mm,mm,mm
500,100,100,30,3,1.5
```

5.9.4 HDF5 Recommendations

For large field datasets, multidimensional arrays, time-dependent image stacks, or voxel data, CSV is not recommended. HDF5 should be used when data are large and/or multidimensional, for example DIC displacement and strain fields, 3D tomography data, voxelated microstructures. time-dependent temperature or moisture fields.

5.9.5 JSON File Standards

JSON files will be used throughout the database to define machine-readable vocabularies, parametrization libraries, validation schemas, metadata structures, and future API interfaces.

In general, CSV files are intended for tabular numerical data such as curves, loading histories, tables, and simple parameter lists, whereas JSON files are intended for structured metadata, parametrization definitions, hierarchical relationships, validation schemas, and machine-readable database definitions.

JSON files should follow the following conventions:

- UTF-8 encoding;
- four-space indentation;
- lowercase underscore-separated field names;
- explicit units whenever numerical quantities are included;
- standardized database identifiers such as `source_id`, `batch_id`, `specimen_id`, `geometry_id`, and `test_id`.

JSON files should conform to the associated JSON Schema validation (*under development*) definitions that will be distributed with the database.

5.10 Controlled Vocabularies

The spreadsheet uses controlled vocabularies (Part 9), primarily through dropdown lists provided in the `Vocab` tab and through accompanying machine-readable JSON vocabulary definition files (*under development*).

Whenever possible, contributors should select existing entries from the predefined vocabularies rather than introducing custom terminology or abbreviations. If an appropriate vocabulary entry does not exist, contributors are encouraged to propose new entries together with concise definitions and, when relevant, associated units or parameter descriptions. The vocabulary system is extensible and will evolve continuously as the database expands.

Vocabulary entries should generally follow lowercase underscore-separated naming conventions.

For example:

```
three_point_bending
chloride_diffusion
carbon_fiber_reinforced_polymer
reversed_cyclic_loading
```

The recommended format for observed mechanisms is:

```
category | mechanism_name
```

This two-level structure separates the broad physical category from the specific observed mechanism, facilitating semantic grouping and database queries.

Examples:

```
strength | unconfined_compressive_failure
fracture | fracture_propagation
transport | chloride_diffusion
fresh | workability
thermal | adiabatic_temperature_rise
durability | alkali_silica_reaction
bond | reinforcement_pullout
```

Multiple mechanisms may be entered in one cell separated by commas. For example:

```
fracture | crack_localization, strength | tensile_failure
```

5.11 Required vs Optional Tabs and Fields

The data collection template contains both required and optional tabs and fields. Required tabs and fields must be completed for the dataset to be accepted into the database. Optional fields should still be completed whenever the corresponding information is available. A field may be

classified as optional because the information is not always reported in historical publications, reports, or experimental records. However, contributors are strongly encouraged to provide all available information to maximize dataset completeness, traceability, and long-term usability.

The table below summarizes the tabs included in the data collection template together with their requirement status and the conditions under which each tab must be completed.

Tab Name	Requirement Status and Conditions
`literature_sources`	Required for all dataset submissions.
`keywords`	Required for all dataset submissions.
`material_batches`	Required for all dataset submissions.
`geometries`	Required for all dataset submissions.
`specimens`	Required for all dataset submissions.
`tests`	Required for all dataset submissions.
`data`	Required for all dataset submissions.
`reinforcement_materials`	Conditionally required if reinforcement materials are present in the specimens.
`reinforcement_layouts`	Conditionally required if reinforcement materials are present in the specimens.
`reinforcement_components`	Conditionally required if reinforcement materials are present in the specimens.
`printing_variables`	Conditionally required if the specimen is additively manufactured.
`devices_and_supports`	Optional but recommended if instrumentation devices, supports, fixtures, actuators, sensors, or boundary-condition components are available.
`loading_histories`	Optional but recommended if loading protocols, cyclic histories, displacement histories, force histories, or time-dependent loading information are reported.

Fields marked as required within each tab are used during automated dataset validation procedures to verify completeness, consistency, identifier relationships, schema compliance, and conformity with the adopted controlled vocabularies. Datasets missing required tabs or required fields may be rejected or flagged for manual review prior to ingestion into the centralized database.

Examples of optional but highly encouraged information include DOI identifiers, detailed device metadata, environmental conditions, extraction dates, notes, supplementary geometry files, and supplementary loading or instrumentation descriptions. Providing this information substantially improves dataset traceability, interoperability, reproducibility, and future reuse for computational modeling, benchmark studies, and AI-assisted data analysis workflows.

Chapter 6

Step-by-Step Manual Submission Workflow - Deprecated

6.1 Create Source Folder

Create one folder for the dataset. The folder must be named using the source identifier:

```
SRC0001
```

Only one dataset should be submitted at a time. Copy the template spreadsheet into the source folder and rename it:

```
SRC0001.xlsx
```

6.2 Create Batch Folders

For each material batch, create a folder inside the source folder:

```
SRC0001/BAT0001
SRC0001/BAT0002
```

Each batch folder must correspond to one row in the `material_batches` tab.

6.3 Create Test Folders

For each test, create a folder inside the corresponding batch folder:

```
SRC0001/BAT0001/TST0001
SRC0001/BAT0001/TST0002
SRC0001/BAT0002/TST0003
```

Use progressive numbering across batches. Do not restart test numbering for each batch.

6.4 Populate the `literature_sources` Tab

The `literature_sources` table records bibliographic and identifying information for the source document, dataset, report, thesis, or experimental campaign from which the submitted data were extracted. The `source_id` is referenced throughout the database hierarchy and serves as the top-level identifier linking all associated batches, specimens, tests, loading histories, devices, and experimental data. In manual submission, use `SRC0001` consistently. During ingestion into the centralized database, local identifiers will be automatically mapped to globally unique database identifiers while preserving the original submitted identifiers for traceability.

Field	Req.	Type	Description
`source_id`	Yes	String	Identifier of the source document or dataset. Use `SRC0001` for manual submission.
`source_type`	No	Vocab	Type of source. Examples include journal article, report, thesis, dataset, conference paper, or technical report.
`peer_reviewed`	No	yes/no	Indicates whether the source was peer reviewed.
`title`	Yes	Text	Full title of the source.
`authors`	Yes	Text	Author list as reported in the source. Use comma separation.
`year`	No	Integer	Publication or release year.
`journal_or_publisher`	No	Text	Journal name, publisher, report agency, hosting institution, or conference proceedings.
`volume`	No	Text	Volume number if available.
`issue`	No	Text	Issue number if available.
`pages`	No	Text	Page range or article number.
`doi`	No	Text	Digital object identifier. Optional only because older sources may not have one.
`url`	No	Text	URL to the source or dataset record.
`institution`	No	Text	Institution associated with a report, thesis, dataset, or experimental campaign.
`notes`	No	Text	Any information useful to identify the source, especially when DOI or URL information is unavailable.

6.5 Populate the `keywords` Tab

The `keywords` table records a **minimum of five keywords** describing the dataset. Keywords are associated with the `source_id` and describe the dataset as a whole rather than one specific specimen or test.

Field	Req.	Type	Description
`source_id`	Yes	String	Identifier of the source associated with the keyword. Use `SRC0001` for manual submission.
`keyword`	Yes	Text	Searchable keyword describing material systems, tests, mechanisms, methods, environmental conditions, or data types associated with the dataset.

6.6 Populate the material_batches Tab

The material_batches table describes each concrete mixture, batch, or material system reported in the source. One row corresponds to one distinct material formulation. Each batch references one source through source_id. The material batch definition is independent of specimen geometry and testing configuration, allowing the same material system to be reused across multiple specimens and tests.

The table supports conventional cement paste, mortar, concrete, fiber-reinforced concrete, cementitious composites, cementitious mixture for additive manufacturing, and other cementitious materials relevant to the database scope. When available, a text file describing the mixing protocol should be included in the corresponding batch folder.

Field	Req.	Type	Units	Description
batch_id	Yes	String	none	Unique identifier for the material batch. Must correspond to the batch folder name.
source_id	Yes	String	none	Source identifier.
mixing_protocol_file_name	No	String	none	Text file describing the mixing, batching, curing, or specimen preparation protocol.
batch_label	No	Text	none	Label used in the original source.
material_class	Yes	Vocab	none	General material class.
cement_type	No	Vocab	none	Cement classification, designation, or product description.
printable	No	yes/no	none	Indicates whether the material is intended for additive manufacturing or extrusion-based fabrication.
self_consolidating	No	yes/no	none	Indicates whether the material is self consolidating.
cement_content_kg_m3	Yes	Number	kg/m^3	Cement mass per unit concrete volume.
water_binder_ratio	Yes	Number	-	Ratio of water mass to total binder mass.
silica_fume_content_kg_m3	No	Number	kg/m^3	Silica fume content.
fly_ash_content_kg_m3	No	Number	kg/m^3	Fly ash content.
fly_ash_class	No	Vocab	-	Fly ash class if reported.
slag_content_kg_m3	No	Number	kg/m^3	Slag cement content.
fine_aggregate_content_kg_m3	No	Number	kg/m^3	Fine aggregate mass per unit concrete volume.
fine_aggregate_dry_density_kg_m3	No	Number	kg/m^3	Dry density of fine aggregate.
coarse_aggregate_content_kg_m3	No	Number	kg/m^3	Coarse aggregate mass per unit concrete volume.
coarse_aggregate_dry_density_kg_m3	No	Number	kg/m^3	Dry density of coarse aggregate.
max_aggregate_size_mm	No	Number	mm	Maximum aggregate size.
aggregate_type	No	Vocab	none	Aggregate mineralogy, lithology, or aggregate description.
superplasticizer_type	No	Text	none	Superplasticizer type or commercial product designation.
superplasticizer_content_kg_m3	No	Number	kg/m^3	Superplasticizer dosage by mass.
water_reducer_type	No	Text	none	Water reducer type or commercial product designation.
water_reducer_content_ml_m3	No	Number	ml/m^3	Water reducer dosage by volume.

Field	Req.	Type	Units	Description
air_entrainment_type	No	Text	none	Air entraining admixture type or product designation.
air_entrainment_content_ml_m3	No	Number	ml/m^3	Air entrainment dosage.
rheology_modifier_type	No	Text	none	Rheology modifying admixture type or product designation.
rheology_modifier_content_kg_m3	No	Number	kg/m^3	Rheology modifier dosage.
hydration_accelerator_type	No	Text	none	Hydration accelerator type or product designation.
hydration_accelerator_content_ml_m3	No	Number	ml/m^3	Hydration accelerator dosage.
fiber_type	No	Vocab	none	Fiber material and fiber form.
fiber_product_name	No	Text	none	Commercial fiber product name if available.
fiber_volume_fraction	No	Number	none or %	Fiber volume fraction. State whether decimal or percent notation is used.
fiber_length_mm	No	Number	mm	Fiber length.
fiber_diameter_mm	No	Number	mm	Fiber diameter.
air_content_percent_volume_concrete	No	Number	%	Air content by concrete volume.
date_of_pouring_yyyy-mm-dd	No	Date	date	Casting date in ISO format.
notes	No	Text	none	Additional information about the mixture, batching procedure, curing protocol, constituent materials, or specimen preparation.

6.7 Populate the reinforcement_materials Tab

The reinforcement_materials tab defines reusable reinforcement material property records that can be referenced by multiple reinforcement components throughout the database. Each entry is uniquely identified by a reinforcement_material_id. The tab depends hierarchically on the literature_sources tab and serves as a parent entity for the reinforcement_components tab.

Field	Req.	Type	Units	Description
reinforcement_material_id	Yes	String	none	Unique reinforcement identifier.
source_id	Yes	String	none	Source identifier.
reinforcement_material_type	No	Vocab	none	Reinforcement material type.
stress_strain_curve_file	No	File name	none	CSV file containing the constitutive response of the reinforcement material.
elastic_modulus_mpa	No	Number	MPa	Elastic modulus of reinforcement material.
yield_strength_mpa	No	Number	MPa	Yield strength of reinforcement material.
rupture_strength_mpa	No	Number	MPa	Ultimate or rupture strength of reinforcement material.
rupture_strain	No	Number	none	Rupture strain of reinforcement material.
notes	No	Text	none	Additional reinforcement information.

The stress_strain_curve_file field identifies a file containing the constitutive response of the

reinforcement material. The file includes stress-strain data describing the mechanical behavior of the reinforcement material under uniaxial loading. The data may represent experimental measurements, manufacturer-provided constitutive curves, or idealized constitutive relationships used in numerical simulations.

The file must contain two header rows followed by as many rows as needed to define the stress-strain curve:

1. variable names separated by commas;
2. variable units separated by commas;
3. stress-strain data rows separated by commas.

General format:

```
strain,stress
-,MPa
e0,s0
e1,s1
...
en,sn
```

Example:

```
strain,stress
-,MPa
0.0000,0
0.0010,200
0.0020,420
0.0500,550
...
```

6.8 Populate the geometries Tab

The geometries tab stores information describing the specimen geometry independently of the material batch, reinforcement configuration, specimen definition, and experimental test. Its main purpose is to define reusable geometric representations of specimens using standardized parametrizations provided in the geometry parametrization library. Each geometry entry defines the physical shape and dimensions of a specimen configuration that may later be associated with one or more specimen definitions and experimental tests.

Only geometries currently defined in the geometry parametrization library are presently supported. When a geometry type cannot be adequately represented using the existing library, contributors are encouraged to propose extensions together with the associated parameter definitions and coordinate conventions.

Field	Req.	Type	Units	Description
geometry_id	Yes	String	none	Unique geometry identifier.
source_id	Yes	String	none	Source identifier.

Field	Req.	Type	Units	Description
specimen_geometry_type	Yes	Vocab	none	Geometry type from the geometry parametrization library.
geometry_file_name	Yes	File name	none	CSV or JSON file containing the geometry parameter names, units, and parameter values according to the geometry parametrization library.
notes	No	Text	none	Additional geometry information.

For each geometry row, create a geometry CSV or JSON file and place it in the corresponding source folder associated with the dataset.

The geometry parameters, parameter names, coordinate conventions, and required dimensions are defined in the geometry parametrization library. The current library includes standardized parametrizations for prisms, cubes, cylinders, beams, notched beams, panels, walls, plates, L-shaped specimens, dog-bone specimens, and other common experimental configurations.

The CSV file must contain exactly three rows: (1) parameter names separated by commas; (2) parameter units separated by commas; (3) parameter values separated by commas.

CSV example for notched_prism:

```
prism_length,prism_depth,prism_thickness,notch_offset,
notch_depth,notch_width,notch_tip_radius
mm,mm,mm,mm,mm,mm,mm
500.0,100.0,100.0,0.0,30.0,3.0,1.5
```

The JSON file must conform to the specifications defined in geometry_schema.json.

Example for prism

```
{
  "geometry_id": "GEO0001",
  "source_id": "SRC0001",
  "specimen_geometry_type": "prism",
  "coordinate_system": {
    "origin": "corner point shown in the prism geometry scheme",
    "x_axis": "longitudinal direction along the prism length",
    "y_axis": "transverse direction along the prism thickness",
    "z_axis": "vertical direction along the prism depth"
  },
  "parameters": {
    "length": {
      "value": 500.0,
      "unit": "mm"
    },
    "thickness": {
      "value": 100.0,
      "unit": "mm"
```

```
    },
    "depth": {
      "value": 100.0,
      "unit": "mm"
    }
  }
}
```

Contributors should ensure that all geometric parameters are reported consistently with the coordinate system and parameter definitions specified in the geometry parametrization library. A drawing, CAD file, annotated figure, or image describing the specimen geometry may also be included in the dataset when available through the `geometry_file_name` field. This supplementary file does not replace the required parametric geometry definition.

6.9 Populate the `printing_variables` Tab

under development

6.10 Populate the `reinforcement_layouts` Tab

The `reinforcement_layouts` tab defines reinforcement configurations associated with specimen geometries but reusable for the definition of multiple specimens. Each entry in this tab represents a complete reinforcement layout composed of one or more reinforcement components subsequently defined in the `reinforcement_components` tab. A single reinforcement layout may therefore be associated with multiple entries in the `reinforcement_components` tab, each describing one constituent reinforcement component together with its associated reinforcement material properties.

The tab supports three principal reinforcement categories:

- longitudinal tensile and flexural reinforcement;
- transverse reinforcement systems such as stirrups, ties, and confinement reinforcement;
- externally bonded or externally attached reinforcement systems, including externally bonded FRP laminates, sheets, and mechanically attached strengthening systems.

Field	Req.	Type	Units	Description
`reinforcement_layout_id`	Yes	String	none	Unique specimen identifier.
`source_id`	Yes	String	none	Source identifier.
`geometry_id`	Yes	String	none	Geometry identifier referencing an entry in the `Geometry` tab.
`longitudinal_reinforcement`	Yes	yes/no	none	Indicates presence of longitudinal reinforcement.
`transverse_reinforcement`	Yes	yes/no	none	Indicates presence of transverse reinforcement such as stirrups or ties.

Field	Req.	Type	Units	Description
external_reinforcement	Yes	yes/no	none	Indicates presence of externally bonded or externally attached reinforcement systems.
reinforcement_file_name	No	File name	none	Optional drawing, CAD file, schematic, annotated image, or supplementary reinforcement description file.
notes	No	Text	none	Additional information describing the reinforcement configuration, reinforcement arrangement, fabrication procedure, etcetera.

6.11 Populate the reinforcement_components Tab

The reinforcement_components tab defines reusable reinforcement components associated with reinforcement layouts. Each entry represents one reinforcement component, such as a rebar layer, stirrup set, or external reinforcement system, and is uniquely identified by a reinforcement_component_id using the format REF0001.

The tab depends hierarchically on the literature_sources, reinforcement_materials, and reinforcement_layouts tabs. Each reinforcement component combines geometric and placement information with the corresponding reinforcement material properties referenced through the reinforcement_material_id. Multiple reinforcement components may be associated with the same reinforcement layout to define complete reinforcement configurations consistently with the coordinate system and parametrization of the associated specimen geometry.

For each reinforcement entry, all geometric parameters, coordinate definitions, and orientation conventions must follow the parametrizations defined in the reinforcement parametrization library.

Populate cells A-G with general information about the reinforcement. Populate cells H-Q if the reinforcement consists of tensile reinforcement, stirrups, or ties. Populate cells R-AL if the reinforcement consists of external reinforcement systems such as FRP wraps, plates, sheets, or externally bonded laminates.

Field	Req.	Type	Units	Description
General Reinforcement Information: Cells A-G				
reinforcement_component_id	Yes	String	none	Unique reinforcement identifier.
source_id	Yes	String	none	Source identifier.
reinforcement_material_id	Yes	String	none	Associated reinforcement material identifier.
reinforcement_layout_id	Yes	String	none	Associated reinforcement layout identifier.
notes	No	Text	none	Additional reinforcement information.
reinforcement_type	Yes	Vocab	none	Reinforcement type from the controlled vocabulary.
reinforcement_orientation	Yes	Vocab	none	Principal reinforcement orientation relative to the specimen coordinate system.

Field	Req.	Type	Units	Description
Tensile Reinforcement, Stirrups, and Ties: Cells H-Q				
number_of_reinf_elements	No	Integer	none	Number of reinforcement bars, stirrups, or ties represented by the entry.
bar_designation	No	Vocab	none	Standard reinforcement bar designation if applicable. If included, it may pre-populate the nominal diameter and area fields.
bar_nominal_diameter_mm	No	Number	mm	Nominal reinforcement diameter. It is prepopulated if a standard bar designation is selected. It requires input if "other" is selected in bar designation.
bar_nominal_area_mm2	No	Number	mm^2	Nominal reinforcement cross-sectional area. It is prepopulated if a standard bar designation is selected. It requires input if "other" is selected in bar designation.
layer_depth_mm	No	Number	mm	Depth of the reinforcement layer measured in the associated system of reference. Leave empty for stirrups and ties when not applicable.
spacing_mm	No	Number	mm	Center-to-center spacing between reinforcement elements. For longitudinal reinforcement this represents spacing between adjacent bars in a layer. For stirrups and ties this represents longitudinal spacing.
horizontal_cover_mm	No	Number	mm	Horizontal concrete cover.
vertical_cover_mm	No	Number	mm	Vertical concrete cover.
begin_distance_mm	No	Number	mm	Distance from the specimen origin to the beginning of the reinforcement system.
end_distance_mm	No	Number	mm	Distance from the specimen origin to the end of the reinforcement system.
External Reinforcement Systems: Cells R-AL				
reinforcement_centroid_x_mm	No	Number	mm	x-coordinate of the external reinforcement centroid.
reinforcement_centroid_y_mm	No	Number	mm	y-coordinate of the external reinforcement centroid.
reinforcement_centroid_z_mm	No	Number	mm	z-coordinate of the external reinforcement centroid.

Field	Req.	Type	Units	Description
reinforcement_width_mm	No	Number	mm	Width of the external reinforcement system.
reinforcement_length_mm	No	Number	mm	Length of the external reinforcement system.
reinforcement_thickness_mm	No	Number	mm	Thickness of the external reinforcement system.
reinforcement_angle_deg	No	Number	deg	Orientation angle of the external reinforcement with respect to the structural member axis.
adhesive_type	No	Vocab	none	Adhesive or bonding material type.
adhesive_thickness_mm	No	Number	mm	Thickness of the adhesive layer.
adhesive_modulus_mpa	No	Number	MPa	Elastic modulus of the adhesive material.
adhesive_strength_mpa	No	Number	MPa	Adhesive strength.
adhesive_failure_strain	No	Number	none	Adhesive failure strain.
anchorage_presence	No	yes/no	none	Indicates whether anchorage systems are present.
anchorage_type	No	Vocab	none	Type of anchorage system.
anchorage_size_mm	No	Number	mm	Characteristic anchorage size or diameter.
anchorage_location_x_mm	No	Number	mm	x-coordinate of the anchorage location.
anchorage_location_y_mm	No	Number	mm	y-coordinate of the anchorage location.
anchorage_location_z_mm	No	Number	mm	z-coordinate of the anchorage location.
anchorage_modulus_mpa	No	Number	MPa	Elastic modulus of the anchorage material.
anchorage_strength_mpa	No	Number	MPa	Strength of the anchorage material.
anchorage_failure_strain	No	Number	none	Failure strain of the anchorage material.

6.12 Populate the specimens Tab

The specimens tab defines the physical specimens or specimen families used in the experimental program. Each entry is uniquely identified by a specimen_id using the format SPC0001 and combines the corresponding material batch, geometry definition, and reinforcement and printing variables (if relevants) into a single specimen definition.

The tab depends hierarchically on the literature_sources, material_batches, geometries, printing_variables, and reinforcement_layouts tabs. It serves as the parent entity for the tests tab, enabling multiple experimental tests to be associated with the same specimen definition.

Field	Req.	Type	Units	Description
specimen_id	Yes	String	none	Unique specimen identifier.

Field	Req.	Type	Units	Description
source_id	Yes	String	none	Source identifier.
batch_id	Yes	String	none	Associated material batch identifier.
geometry_id	Yes	String	none	Geometry identifier referencing an entry in the Geometry tab.
reinforcement_layout_id	Yes	String	none	Geometry identifier referencing an entry in the Geometry tab.
printing_variables_id	Yes	String	none	Geometry identifier referencing an entry in the Geometry tab.
structural_form	No	Vocab	none	Classification describing the overall structural configuration, load-carrying mechanism, or structural member category of the specimen.
notes	No	Text	none	Additional information describing not included elsewhere in the tab.

6.13 Populate the tests Tab

The Tests tab records information about each experimental test performed on a specimen or structural component. Its purpose is to provide a standardized description of the testing procedure, environmental conditions, loading configuration, and observed physical response associated with the experiment.

The tab includes information about the test type, specimen age at testing, curing history, thermal and hygral boundary conditions, environmental temperature and humidity, and observed physical mechanisms or failure modes. The tab also supports optional parametrization of the test setup through dedicated parameter files defined according to the test parametrization library.

Each test entry references the corresponding source_id, batch_id, and specimen_id, thereby linking the experimental test to the material system and specimen configuration defined elsewhere in the database hierarchy. This separation between specimen definitions and tests minimizes redundancy and allows the same specimen configuration to be reused across multiple experimental conditions or loading protocols.

The tab additionally supports links to auxiliary files such as test setup visualizations, test protocols, environmental histories, and other supporting documentation necessary for interpretation and reproducibility of the experimental results.

Each test should also be associated with the relevant entries in the Devices_and_Supports, Loading_Histories, and Data tabs to ensure complete documentation of instrumentation, loading protocols, and collected measurements.

Field	Req.	Type	Units	Description
test_id	Yes	String	none	Unique test identifier. Must correspond to the test folder name.
source_id	Yes	String	none	Source identifier.
specimen_id	Yes	String	none	Specimen identifier referencing the corresponding entry in the Specimens tab.

Field	Req.	Type	Units	Description
number_of_repeats	Yes	String	none	Record the number of same tests done on the same specimen.
test_setup_visualization_file_name	No	File name	none	Image, drawing, photograph, CAD rendering, or diagram illustrating the experimental setup.
test_protocol_file_name	No	File name	none	File containing the experimental protocol, loading sequence, instrumentation details, or testing procedure.
test_type	Yes	Vocab	none	Type of experimental test from the controlled vocabulary.
test_parameters_file_name	No	File name	none	File containing the test setup parameters defined according to the test parametrization library.
main_observed_mechanisms	Yes	Vocab	none	Main observed physical mechanisms or failure modes. Use `category | mechanism` format. Multiple mechanisms may be separated by commas.
age_days	No	Number	days	Specimen age at testing.
curing_condition	No	Vocab	none	Curing condition before testing.
hygral_boundary_condition	No	Vocab	none	Moisture boundary condition during testing.
thermal_boundary_condition	No	Vocab	none	Thermal boundary condition during testing.
initial_env_temperature_C	No	Number	°C	Initial environmental temperature.
env_temperature_history_file_name	No	File name	none	File containing the environmental temperature history recorded during the test.
initial_env_relative_humidity_percent	No	Number	%	Initial environmental relative humidity.
env_relative_humidity_history_file_name	No	File name	none	File containing the environmental relative humidity history recorded during the test.
test_date_yyyy-mm-dd	No	Date	date	Date of testing in ISO format.
notes	No	Text	none	Additional information regarding the test setup, loading configuration, specimen condition, environmental exposure, or interpretation of the experimental results.

For each test entry, create the corresponding files listed below and place them in the associated test folder.

- `test_setup_visualization_file_name` identifies an optional, but strongly recommended, image file illustrating the experimental setup. The file may be a photograph, schematic, CAD drawing, annotated figure, or diagram showing specimen placement, supports, loading devices, instrumentation, and relevant boundary conditions.

- `test_protocol_file_name` identifies an optional file containing the experimental protocol and supporting documentation. The file may include testing standards, loading procedures, instrumentation details, calibration information, specimen preparation procedures, data acquisition protocols, or safety instructions.

- `test_parameters_file_name` identifies a file containing the parameters defining the test setup according to the parametrization specified in the test parametrization library. This file is required whenever the selected test type is parametrized in the library.

  The file must contain exactly three rows:

  1. parameter names separated by commas;
  2. parameter units separated by commas;
  3. parameter values separated by commas.

  Example for a `three_point_bending` test:

  ```
  span
  mm
  5000.00
  ```

- `env_temperature_history_file_name` identifies a file containing the time history of the environmental temperature recorded during the test. The file includes time-temperature data describing variations in ambient or chamber temperature throughout the experimental procedure.

  The file must contain two header rows followed by as many rows as needed to define the temperature history:

  1. variable names separated by commas;
  2. variable units separated by commas;
  3. time-history data rows separated by commas.

  General format:

  ```
  time,temperature
  s,C
  0,T0
  t1,T1
  ...
  tn,Tn
  ```

  Example:

  ```
  time,temperature
  s,C
  0,20
  3600,20
  7200,35
  10800,50
  ```

- `env_relative_humidity_history_file_name` identifies a file containing the time history of the environmental relative humidity recorded during the test. The file includes time-humidity data describing variations in ambient or chamber relative humidity throughout the experimental procedure.

  The file must contain two header rows followed by as many rows as needed to define the relative humidity history:

  1. variable names separated by commas;
  2. variable units separated by commas;
  3. time-history data rows separated by commas.

  General format:

```
time,relative_humidity
s,percent
0,RH0
t1,RH1
...
tn,RHn
```

  Example:

```
time,relative_humidity
s,percent
0,50
3600,50
7200,65
10800,80
```

6.14 Populate the `devices_and_supports` Tab

The `Devices_and_Supports` tab records information about the devices, supports, fixtures, and instrumentation associated with an experimental test. Its purpose is to document the relevant components of the experimental setup, including their type, location, orientation, and defining parameters.

Each device or support entry references the corresponding `source_id`, `batch_id`, `geometry_id`,and `test_id`. Device and support coordinates are defined within the Cartesian coordinate system associated with the corresponding specimen geometry.

Field	Req.	Type	Units	Description
`device_id`	Yes	String	none	Unique identifier for the device or support entry.
`source_id`	Yes	String	none	Source identifier.
`test_id`	Yes	String	none	Associated experimental test.

Field	Req.	Type	Units	Description
device_type	Yes	Vocab	none	Type of device, support, fixture, or instrumentation component. Examples include extensometer, LVDT, CMOD gauge, DIC system, camera, load cell, actuator, roller support, hinge support, or loading platen.
x_location_mm	No	Number	mm	x-coordinate of the device or support location.
y_location_mm	No	Number	mm	y-coordinate of the device or support location.
z_location_mm	No	Number	mm	z-coordinate of the device or support location.
orientation	No	Vocab	none	x, y, z orientation of the device or support.
device_parameters_file	No	File name	none	CSV file including parametrs characterizing the device as per the device parametrization file. Examples of parameters include gage length, support diameter, platen width, camera resolution, or sampling frequency. For more complex parameters data structure a JSON file may be required. See devices_and_supports library.
notes	No	Text	none	Additional information describing not included elsewhere in the tab.

For each devices_and_supports row, a CSV file may be crated and placed in the corresponding test folder if the device or support is included in the device and support parametrization library. The file must contain three rows:

1. parameter names separated by commas
2. parameter units separated by commas
3. parameter values separated by commas

The devices_and_supports parameters and their name are defined in the devise and support library.

Example for roller_support:

```
length,diameter
mm,mm
10,50
```

For more complex description of a support or device contributors should use the JSON schema contained in the schema specified devices_and_supports_library.json.

6.15 Populate the loading_histories Tab

The Loading_Histories tab records information about the loading protocols and loading time histories associated with an experimental test. Its purpose is to provide a standardized description of the loading conditions applied during the experiment, including loading mode, control variables, loading rates, time intervals, and associated loading-history files.

Each loading history entry references the corresponding source_id, batch_id, geometry_id, test_id, and, when relevant, the associated device_id used to apply or measure the loading history.

The tab supports a broad range of loading conditions including monotonic loading, cyclic loading, fatigue loading, creep loading, thermal loading, hygral loading, and multiphysics loading histories.

Field	Req.	Type	Units	Description
loading_id	Yes	String	none	Unique loading history identifier.
source_id	Yes	String	none	Source identifier.
test_id	Yes	String	none	Associated experimental test.
device_id	No	String	none	Associated loading actuator, device, or instrumentation component if applicable.
loading_mode	Yes	Vocab	none	Type of loading protocol or loading mode. Examples include monotonic, cyclic, fatigue, creep, relaxation, thermal, or hygral loading.
loading_control_variable	Yes	Vocab	none	Primary controlled quantity defining the loading protocol. Examples include force, displacement, strain, stress, temperature, pressure, or humidity.
initial_control_variable_rate	No	Number	various	Initial loading rate associated with the control variable.
control_variable_rate_units	No	Text	various	Units associated with the control-variable loading rate. Examples include kN/s, mm/min, strain/s, or °C/h.
begin_time_sec	No	Number	s	Initial time associated with the loading history.
end_time_sec	No	Number	s	Final time associated with the loading history.
loading_history_file_name	No	File name	none	CSV file containing the loading time history.
notes	No	Text	none	Additional information describing not included elsewhere in the tab.

The loading_history_file_name identifies a file containing the loading history associated with the experimental test. The file may include force, displacement, strain, temperature, pressure, humidity, or other prescribed or measured loading quantities as a function of time.

The file must contain two header rows followed by as many rows as needed to define the loading history:

1. variable names separated by com
2. variable units separated by commas
3. loading-history data rows separated by commas

General format:

```
time,variable

s,unit

t0,v0

t1,v1

...

tn,vn
```

Example:

```
time,force

s,kN

0,0

10,15

20,30

30,45
```

Multiple loading histories may be associated with the same experimental test. For example, fracture tests on three point bending are often initially controlled via CMOD and later via machine head displacement.

6.16 Populate the data Tab

The data tab records the experimental observations, measurements, processed results, tables, curves, images, and videos associated with an experimental test. Its purpose is to provide a standardized description of all experimental outputs together with their provenance, measurement definitions, associated devices, and corresponding files.

The tab supports multiple categories of experimental data, including:

- scalar quantities;

- curves and time histories;
- tables;
- images and videos.

Each data entry references the corresponding `source_id`, `batch_id`, `geometry_id`, and `test_id`. Whenever relevant, the data entry may also reference one or more devices listed in the `Devices_and_Supports` tab.

Populate cells A-N with general information; cells O-T if data is a scalar; cells U-AA if data is a curve; cells AB-AD if data is an image; and cells AE-AI if data is a video.

Field	Req.	Type	Units	Description
General Information: Cells A-L				
`data_id`	Yes	String	none	Unique data identifier.
`source_id`	Yes	String	none	Source identifier.
`test_id`	Yes	String	none	Associated experimental test.
`test_repeat_id`	Yes	String	none	Associated experimental test repeat.
`notes`	No	Text	none	Additional comments or notes associated with the data entry.
`source_location`	No	Text	none	Location of the data within the original source. Examples include page number, figure number, table number, appendix, or supplementary material reference.
`extraction_method`	No	Vocab	none	Method used to obtain the data. Examples include direct, digitized, AI-assisted, or computed.
`extraction_date_yyyy-mm-dd`	No	Date	date	Date on which the data were extracted or curated.
`curator_first_name`	No	Text	none	First name of the curator responsible for the data entry.
`curator_last_name`	No	Text	none	Last name of the curator responsible for the data entry.
`data_type`	Yes	Vocab	none	Type of data entry. Examples include scalar, curve, table, image, or video.
`number_of_specimens`	No	Integer	none	Number of specimens represented by the data entry.
`specimen_label`	No	Text	none	Original specimen label or identifier used in the source.
Scalar Data: Cells M-R				
`device_id`	No	String	none	Identifier of the device associated with the scalar measurement.

Field	Req.	Type	Units	Description
quantity_reported	Yes	Text	none	Name of the reported scalar quantity.
quantity_reported_type	Yes	Vocab	none	Type or classification of the reported quantity.
quantity_reported_mean	Yes	Num.	various	Mean value of the reported scalar quantity.
quantity_reported_standard_deviation	No	Num.	various	Standard deviation of the reported scalar quantity.
units	No	Text	various	Units associated with the reported quantity.
Curve Data: Cells U-AA				
x_device_id	No	String	none	Identifier of the device associated with the x-axis quantity.
y_device_id	No	String	none	Identifier of the device associated with the y-axis quantity.
x_quantity_reported	Yes	Text	none	Quantity reported along the x-axis.
x_quantity_reported_type	Yes	Vocab	none	Type or classification of the x-axis quantity.
y_quantity_reported	Yes	Text	none	Quantity reported along the y-axis.
y_quantity_reported_type	Yes	Vocab	none	Type or classification of the y-axis quantity.
file_name	Yes	File name	none	CSV file containing the curve or time-history data.
Table Data: Cells: Z-AB				
table_name	Yes	Text	none	Name or title of the table.
table_description	No	Text	none	Description of the table contents.
table_file_name	Yes	File name	none	CSV or spreadsheet file containing the tabulated data.
Image / Video Data: Cells AC-AG				
name	Yes	Text	none	Name or title of the image or video entry.
description	No	Text	none	Description of the image or video contents.
type	No	Vocab	none	Type of image or video. Examples include crack pattern, specimen photograph, test setup image, DIC field, microscopy image, or failure video.
format	No	Text	none	File format of the image or video. Examples include png, jpg, tif, mp4, or avi.

Field	Req.	Type	Units	Description
`filename`	Yes	File name	none	Image or video file associated with the data entry.

The `file_name` field associated with curve data identifies a CSV file containing the curve or time-history data. The file must contain two header rows followed by the data rows.

General format:

```
x_quantity,y_quantity

x_unit,y_unit

x0,y0

x1,y1

...

xn,yn
```

Example:

```
displacement,force

mm,kN

0.00,0.00

0.10,12.50

0.20,25.30

0.30,37.10
```

The `table_file_name` field identifies a CSV or spreadsheet file containing tabulated experimental data. The file may include multidimensional measurements, processed results, or derived quantities associated with the experimental program.

The `filename` field associated with image and video data identifies image or video files documenting the experiment. These files may include photographs, microscopy images, DIC visualizations, crack maps, thermal images, test setup illustrations, or experimental videos relevant to the interpretation of the data.

Chapter 7

AI-Assisted Submission Workflow

under development

Chapter 8

Standardized Libraries

A major challenge in building a machine-readable experimental database for concrete materials and structures is the large variability and heterogeneity of available data.

To address these challenges, the database adopts, when feasible, standardized parametrization schemes for

1. specimen geometries;
2. reinforcement layouts;
3. test setups;
4. devices and supports.

The parametrization schemes for specimen geometries, reinforcement layouts, and test setups are primarily geometric in nature and define standardized coordinate systems, dimensional descriptors, spatial relationships, and geometric parameters required to represent the physical configuration of specimens and experiments. In contrast, the devices and supports parametrization library may additionally include non-geometrical descriptors and constitutive parameters associated with supports, loading devices, instrumentation systems, sensors, actuators, fixtures, and boundary-condition components.

8.1 Reinforcement Geometry Parametrization Library

The Reinforcement Parametrization Library defines the standardized parametric representations used to describe the geometry of reinforcement configurations throughout the database. The library provides canonical parametrizations for longitudinal tensile and flexural reinforcement, transverse reinforcement systems such as stirrups and ties, and externally bonded or externally attached reinforcement systems. Each reinforcement type is described through a consistent set of geometric descriptors, coordinate conventions, orientation definitions, spacing parameters, and material-property descriptors referenced to the Cartesian coordinate system associated with the corresponding specimen geometry.

The adopted geometrical parametrizations are designed to balance simplicity, extensibility, machine readability, and practical usability while capturing the dominant reinforcement characteristics required for structural interpretation, computational modeling, database interoperability, and future AI-assisted workflows. Rather than reproducing every local geometric detail of reinforcement bending shapes, hooks, laps, or anchorage configurations, the library focuses on the principal geometric and mechanical descriptors most relevant for simulation, model calibration, model validation, and systematic comparison across datasets. When more detailed representations are necessary, contributors may supplement the parametrized descriptions with additional drawings, CAD files, annotated figures, photographs, or auxiliary reinforcement description files.

The current reinforcement parametrization schemes and coordinate conventions are described below.

8.1.1 Parametrization of Flexural Reinforcement, Tensile Reinforcement, Stirrups, and Ties

The reinforcement parametrization adopted for flexural reinforcement, tensile reinforcement, stirrups, and ties is based on a standardized layer-based representation intended to describe reinforcement layouts commonly encountered in reinforced concrete specimens and structural members. The same parametrization can be used for longitudinal reinforcement oriented along the x, y, or z directions as well as for transverse reinforcement systems such as stirrups and ties.

Each reinforcement entry represents either:

- one reinforcement layer containing multiple longitudinal bars;
- one set of stirrups or ties repeated at a specified spacing.

The reinforcement layout is defined through the following parameters:

- `number`: number of reinforcement bars, stirrups, or ties represented by the entry;
- `bar_designation`: standard reinforcement designation according to the relevant standard or nomenclature system. The `other` option may be used for non-standard reinforcement;
- `bar_nominal_diameter_mm`: nominal diameter of the reinforcement element;
- `bar_nominal_area_mm2`: nominal cross-sectional area of the reinforcement element;
- `layer_depth_mm`: depth of the reinforcement layer measured from the reference specimen surface. This parameter may be omitted for stirrups and ties;

- `spacing_mm`: center-to-center spacing between reinforcement elements. For longitudinal reinforcement, this parameter defines the spacing between adjacent bars within a layer. For stirrups and ties, it defines the longitudinal spacing between consecutive transverse reinforcement elements;
- `horizontal_cover_mm`: concrete cover measured along the horizontal direction of the specimen coordinate system;
- `vertical_cover_mm`: concrete cover measured along the vertical direction of the specimen coordinate system;
- `begin_distance_mm`: distance from the specimen origin to the beginning of the reinforcement system along the principal reinforcement direction;
- `end_distance_mm`: distance from the specimen origin to the end of the reinforcement system along the principal reinforcement direction.

For tensile and flexural reinforcement, the reinforcement bars are assumed to be equally spaced within the reinforcement layer according to the specified `spacing_mm`. The combination of layer depth, spacing, cover distances, and begin/end distances defines the position and extent of the reinforcement layer within the specimen.

Figure 8.1 illustrates the coordinate conventions and principal geometric descriptors adopted for longitudinal flexural and tensile reinforcement layers.

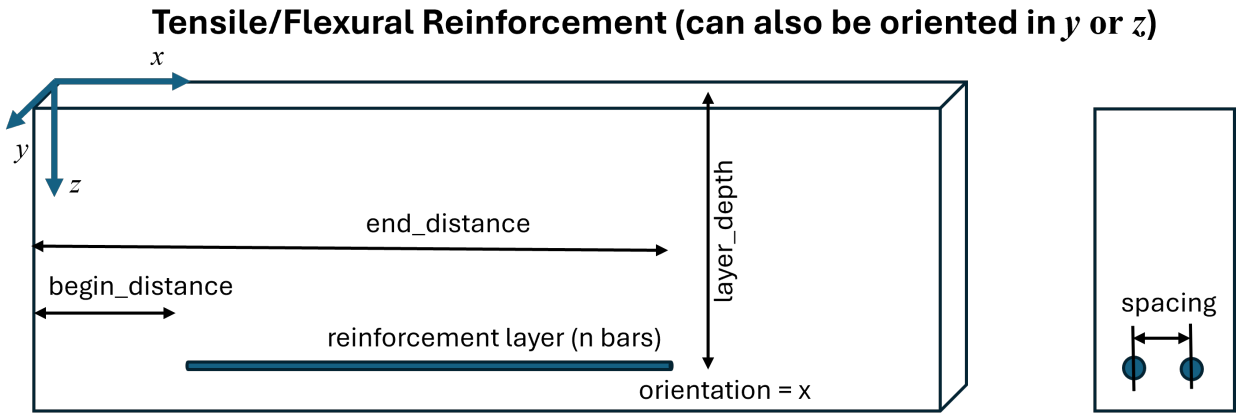


Figure 8.1: Parametrization of longitudinal flexural and tensile reinforcement layers.

For stirrups and ties, the reinforcement entry represents a repeated transverse reinforcement configuration distributed along the member longitudinal direction. The parameter `spacing_mm` defines the center-to-center longitudinal spacing between consecutive stirrups or ties, while `begin_distance_mm` and `end_distance_mm` define the region of the structural member over which the transverse reinforcement is present.

Figure 8.2 illustrates the parametrization adopted for stirrups and ties.

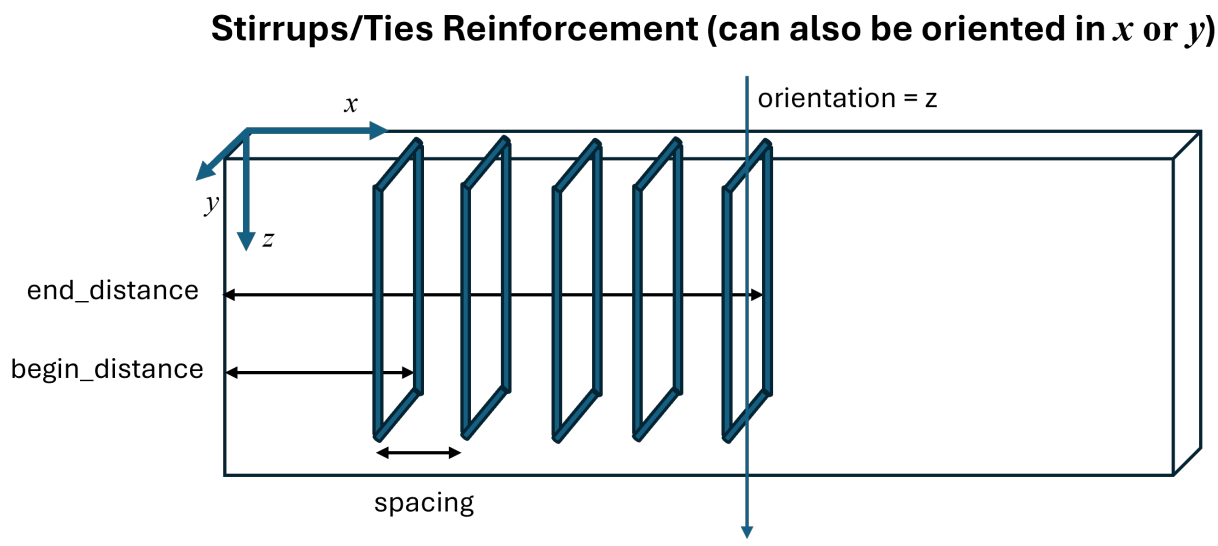


Figure 8.2: Parametrization of stirrups and ties reinforcement systems.

8.2 Parametrization of External Reinforcement

The current parametrization adopted for externally bonded or externally attached reinforcement systems provides a simplified geometric representation intended primarily for machine readability. The parametrization is currently designed for typical externally bonded reinforcement systems such as FRP laminates, steel plates, externally bonded sheets, strips, and similar reinforcement components applied to the surface of structural members.

The current scheme is admittedly limited and focuses only on the dominant geometric descriptors required to identify the reinforcement location, orientation, dimensions, and some limited anchorage characteristics. The parametrization does not attempt to reproduce detailed local geometries, tapering effects, multilayer stacking sequences, corner wrapping details, fiber orientations within composite laminates, or complex anchorage systems. The library is expected to evolve over time, and future versions will include more advanced parametrizations capable of representing a broader range of externally bonded and externally attached reinforcement configurations.

The current parametrization is illustrated in Fig. 8.3. The external reinforcement system is represented through a planar reinforcement region characterized by its centroid location, dimensions, orientation, and optional anchorage information.

The parametrization is defined through the following parameters:

- `reinforcement_centroid_x_mm`: x coordinate of the centroid of the external reinforcement system within the specimen coordinate system;
- `reinforcement_centroid_y_mm`: y coordinate of the centroid of the external reinforcement system within the specimen coordinate system;
- `reinforcement_centroid_z_mm`: z coordinate of the centroid of the external reinforcement system within the specimen coordinate system;
- `reinforcement_width_mm`: width of the external reinforcement region measured perpendicular to the principal reinforcement direction;

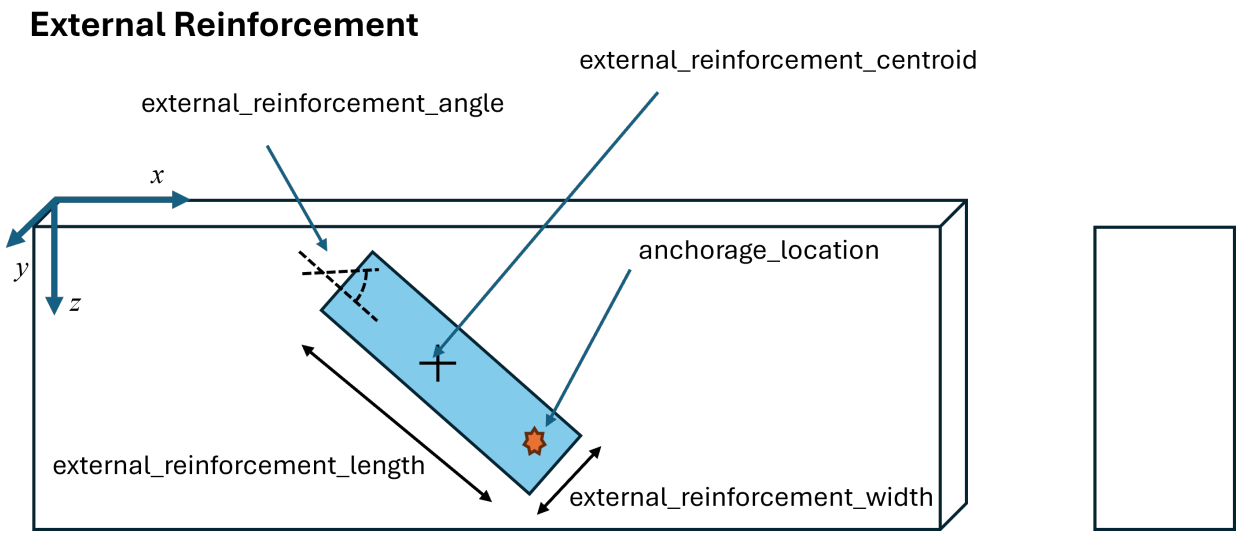


Figure 8.3: Parametrization of external reinforcement systems.

- `reinforcement_length_mm`: length of the external reinforcement region measured along the principal reinforcement direction;
- `reinforcement_thickness_mm`: thickness of the external reinforcement system;
- `reinforcement_angle_deg`: orientation angle of the external reinforcement measured relative to the reference structural axis shown in Fig. 8.3;
- `anchorage_size_mm`: characteristic anchorage dimension or diameter;
- `anchorage_location_x_mm`: x coordinate of the anchorage location;
- `anchorage_location_y_mm`: y coordinate of the anchorage location;
- `anchorage_location_z_mm`: z coordinate of the anchorage location;

The reinforcement centroid coordinates define the global position of the external reinforcement system within the specimen coordinate system. The parameters `reinforcement_length_mm`, `reinforcement_width_mm`, and `reinforcement_thickness_mm` define the overall dimensions of the reinforcement region, while `reinforcement_angle_deg` defines its orientation relative to the structural member axes.

The adhesive-related parameters define the main mechanical characteristics of the bonding layer between the external reinforcement and the structural substrate. The anchorage-related parameters are optional and provide a simplified representation of supplemental anchorage systems used to improve force transfer or delay debonding failure.

When the current parametrization is insufficient to represent complex externally bonded reinforcement systems, contributors are encouraged to supplement the database entry with additional drawings, CAD files, annotated figures, photographs, laminate stacking descriptions, fiber-orientation specifications, or auxiliary documentation describing the reinforcement configuration in greater detail.

8.3 Test Geometry Parametrization Library

The Test Geometry Parametrization Library defines the standardized parametric representations used to describe the geometry of experimental test setups throughout the database. Its primary purpose is to capture the geometric characteristics of the loading and boundary-condition configuration that are required for correct interpretation, comparison, and computational reproduction of experimental results. In many experimental tests, the interpretation of measured quantities depends not only on the specimen geometry, but also on the geometry of the test setup itself. For example, in modulus-of-rupture tests, the bending span is required, together with the specimen geometry, to compute the tensile strength from the measured peak load. Similarly, support spacing, loading-point locations, and other geometric descriptors may directly influence stress distributions, boundary conditions, failure mechanisms, and measured responses. The library therefore provides standardized parameter definitions and geometric descriptors for common experimental setups such as three-point bending, four-point bending, direct tension, compression, shear, pullout, torsion, and other test configurations.

8.3.1 `three_point_bending_test`

The `three_point_bending_test` parametrization defines the geometry of a three-point bending test setup through the following parameter (see Fig. 8.4):

- `bending_span`: center-to-center distance between the two supports in the bending test configuration. The bending span defines the effective loaded span of the specimen and is a fundamental parameter for computing flexural stresses, modulus of rupture, stiffness, fracture energy, and other derived bending quantities.

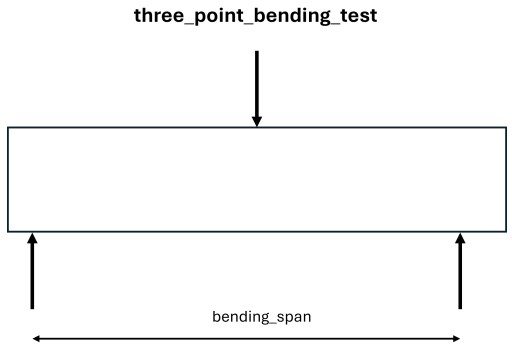


Figure 8.4: Parametrization of external reinforcement systems.

8.3.2 `four_point_bending_test` and `modulus_of_rupture_test`

The `four_point_bending_test` (also applies to `modulus_of_rupture.text`) parametrization defines the geometry of a four-point bending test setup through the following parameters (see Fig. 8.5):

- `bending_span`: center-to-center distance between the two supports in the bending test configuration. The bending span defines the effective loaded span of the specimen and is a fundamental parameter for computing flexural stresses, stiffness, fracture energy, and other derived bending quantities.

- `shear_span`: distance between a support and the nearest loading point measured along the longitudinal axis of the specimen. In symmetric four-point bending configurations, the shear span is equal to one-half of the difference between the bending span and the loading span. The shear span strongly influences the internal shear-force distribution, cracking patterns, failure mode, and the relative contribution of flexural and shear mechanisms.

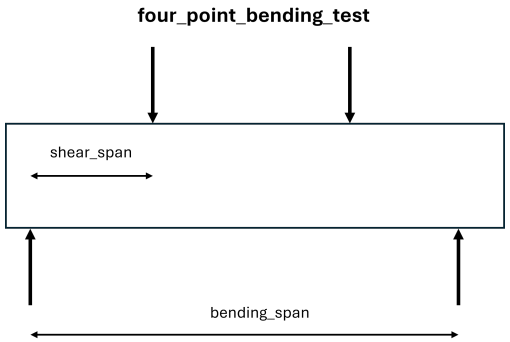


Figure 8.5: Parametrization of the four-point bending test setup.

8.3.3 `axial_test`

The `axial_test` parametrization defines the geometry of an axial loading test in which the applied load is offset from the centroidal axis of the specimen, thereby introducing combined axial force and bending effects (see Fig. 8.6).

- `eccentricity`: perpendicular distance between the line of action of the applied axial load and the centroidal axis of the specimen cross section. The eccentricity governs the magnitude of the induced bending moment.

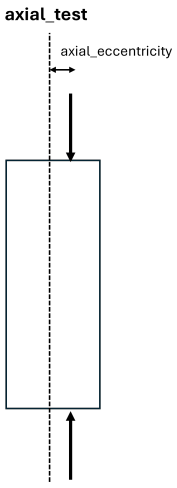


Figure 8.6: Parametrization of the axial loading test with eccentricity.

8.4 Devices & Supports Parametrization Library

The Devices & Supports Parametrization Library defines the standardized parametric representations used to describe the experimental devices, supports, fixtures, loading components, and instrumentation systems associated with experimental tests throughout the database. Its primary purpose is to provide structured machine-readable descriptions of the key characteristics of experimental components that may influence the interpretation, accuracy, reproducibility, and computational representation of test results. The library will include standardized parameter definitions for a broad range of devices and supports such as LVDTs, extensometers, CMOD gauges, load cells, actuators, loading platens, rollers, hinges, soft pads, clamps, cameras, and DIC systems. Depending on the device type, the parametrization may include geometric descriptors, spatial locations, orientations, measurement directions, gage lengths, support dimensions, platen sizes, load-cell capacities, sensor resolutions, sampling frequencies, stiffness properties, constitutive properties, and other relevant parameters required to characterize the experimental setup. Additional parameters and more advanced parametrization may be incorporated in future extensions of the devices and supports parametrization library distributed through the accompanying `devices_and_supports_library.json` file (*under development*).

8.4.1 `pad_support`

The `pad_support` parametrization defines deformable pad supports used to model compliant boundary conditions in experimental setups (Fig. 8.7). The parametrization is intended to represent elastomeric pads, neoprene supports, soft interface layers, wooden strips, or other deformable support materials commonly adopted to reduce stress concentrations, accommodate local rotations, or provide distributed contact conditions between the specimen and the supporting structure.

All coordinates and dimensions are defined within the right-handed Cartesian coordinate system associated with the corresponding specimen geometry. The support location is specified through the coordinates `x_location_mm`, `y_location_mm`, and `z_location_mm`, while the parameter `orientation` defines the principal support direction relative to the specimen coordinate system.

The deformable pad is defined through the following parameters:

- `depth`: pad dimension measured along the axis defined by the `orientation` parameter and orthogonal to the contact surface;
- `thickness`: pad thickness measured in-plane to the contact surface;
- `length`: pad dimension measured in-plane to the contact surface;
- `material`: optional description of the pad material;
- `elastic_modulus`: elastic modulus of the pad material;
- `yielding_stress`: optional yield stress of the pad material;
- `stress_vs_strain_curve`: optional embedded constitutive stress-strain response of the pad material represented through pairs of strain and stress values.

Additional parameters and more advanced support descriptions may be incorporated in future extensions of the devices and supports parametrization library distributed through the accompanying `devices_and_supports_library.json` file.

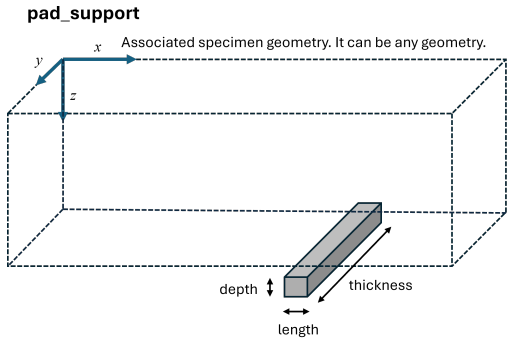


Figure 8.7: Parametrization of deformable pad supports.

8.4.2 `roller_support`

The `roller_support` parametrization defines cylindrical roller supports used to constrain only deflection in the direction of the sliding plane (Fig. 8.8). Roller supports are commonly adopted in flexural, fracture, and structural testing to provide translational freedom along the sliding plane and minimizing restraint forces and parasitic moments at the specimen boundaries.

All coordinates and dimensions are defined within the right-handed Cartesian coordinate system associated with the corresponding specimen geometry. The support location is specified through the coordinates `x_location_mm`, `y_location_mm`, and `z_location_mm`, while the parameter `orientation` defines the principal axis of the cylindrical roller relative to the specimen coordinate system.

The cylindrical roller support is defined through the following parameters:

- `length`: total roller length measured along the roller axis defined by the `orientation` parameter;
- `diameter`: diameter of the cylindrical roller;
- `material`: optional description of the roller material;
- `elastic_modulus`: optional elastic modulus of the roller material;
- `yielding_stress`: optional yield stress of the roller material;
- `surface_roughness`: optional characteristic surface roughness of the roller contact surface.

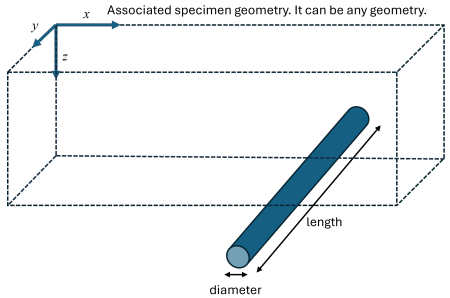


Figure 8.8: Parametrization of cylindrical roller supports.

Chapter 9

Controlled Vocabulary Reference

This part summarizes the controlled vocabularies adopted throughout the database. The controlled vocabulary entries include both the standardized fields used in the data-collection spreadsheet tabs and the predefined lists contained in the `Vocab` tab, which define the drop-down menus used across the various spreadsheet tabs. These vocabularies are further complemented by accompanying machine-readable JSON vocabulary definition files (*currently under development*).

- `identifiers`
- `literature_sources_tab`
  - `source_type_list`
- `material_batches_tab`
  - `material_class_list`
  - `cement_type_list`
  - `aggregate_type_list`
  - `superplasticizer_type_list`
  - `water_reducer_type_list`
  - `air_entrainment_type_list`
  - `rheology_modifier_type_list`
  - `hydration_accelerator_type_list`
  - `fiber_type_list`
- `reinforcement_materials_tab`
  - `reinforcement_material_type_list`
- `geometries_tab`
  - `specimen_geometry_type_list`
- `reinforcement_components_tab`
  - `reinforcement_material_type_list`
  - `reinforcement_type_list`

  - bar_designation_list
  - adhesive_type_list
  - anchorage_type_list
- reinforcement_layouts_tab
- specimens_tab
  - structural_form_type_list
- test_fields_tab
  - test_type_list
  - observed_mechanisms_list
  - curing_conditions_list
  - hygral_boundary_conditions_list
  - thermal_boundary_conditions_list
- devices_and_supports_fields_tab
  - devices_and_supports_list
- loading_history_fields_tab
  - loading_modes_list
  - loading_control_variable_list
- data_fields_tab
  - extraction_methods_list
  - data_type_list
  - quantity_reported_list
  - quantity_reported_type_list

9.1 Identifiers

The following table defines the identifiers used in the database. The machine-readable definitions of these vocabulary entries are provided in `identifiers.json`. The database adopts standardized identifiers to uniquely define and reference the main entities contained in the hierarchical data structure. Each identifier follows a fixed alphanumeric format composed of a prefix indicating the entity type and a progressive numerical index. These identifiers are used throughout the spreadsheet, JSON files, validation schemas, and file organization structure.

Identifier	Format	Definition
source_id	SRC0001	Unique identifier of a literature source, report, thesis, dataset, or experimental campaign representing the top-level entity in the database hierarchy.

Identifier	Format	Definition
batch_id	BAT0001	Unique identifier of a material mixture, material batch, or material system definition.
reinforcement_material_id	RMT0001	Unique identifier of a reusable reinforcement material definition containing the material properties associated with reinforcement systems.
geometry_id	GEO0001	Unique identifier of a reusable specimen geometry definition based on the geometry parametrization library.
printing_id	PRI0001	Unique identifier of a reusable definition of additive manufacturing or printing process variables.
reinforcement_layout_id	RFL0001	Unique identifier of a reusable reinforcement layout associated with a specimen geometry and composed of one or more reinforcement components.
reinforcement_component_id	REF0001	Unique identifier of an individual reinforcement component, such as a rebar layer, stirrup set, tie set, or external reinforcement system, associated with a reinforcement layout.
specimen_id	SPC0001	Unique identifier of a physical specimen or specimen family defined by the combination of material batch, geometry, optional reinforcement layout, and optional printing variables.
test_id	TST0001	Unique identifier of an experimental test performed on a specimen.
device_id	DEV0001	Unique identifier of a support, actuator, sensor, fixture, instrumentation component, or measurement device associated with a test.
loading_id	LDG0001	Unique identifier of a loading protocol or loading history associated with a test and, when applicable, with a specific device.
data_id	DAT0001	Unique identifier of an experimental observation or dataset, including scalar quantities, curves, tables, images, videos, and multidimensional field data.

9.2 Literature Source Tab Fields

The following table defines the fields used in the literature_sources tab. The machine-readable definitions of these vocabulary entries are provided in literature_source_tab_fields.json.

Field Name	Definition
source_type	Classification of the source document type.
peer_reviewed	Boolean field indicating whether the source underwent formal peer review prior to publication.

Field Name	Definition
title	Full title of the publication.
authors	List of authors associated with the source document.
year	Year of publication or public release of the source document.
journal_or_publisher	Name of the journal, conference proceedings, publishing organization, institution, or publisher associated with the source.
volume	Journal or proceedings volume number when applicable.
issue	Journal or proceedings issue number when applicable.
pages	Page range associated with the source document when applicable.
doi	Digital Object Identifier associated with the source document.
url	Web address or online repository link associated with the source document or dataset.
institution	Institution, laboratory, university, company, agency, or organization associated with the source document or experimental program.

9.2.1 Literature Source Types

The following table defines the controlled vocabulary for the source_type field used in the literature_sources tab. The machine-readable definitions of these vocabulary entries are provided in source_types.json.

Term	Definition
book_chapter	A chapter published within an edited book or proceedings volume.
conference_paper	A paper published in conference proceedings or presented at a scientific or engineering conference.
dataset	A standalone curated dataset, database, or data repository entry.
journal_article	An article published in a scientific, technical, or professional journal.
phd_thesis	A doctoral dissertation or PhD thesis submitted as part of a doctoral degree program.
report	A technical, institutional, governmental, industrial, or research report.
other	A source type that does not fall within the predefined categories.

9.3 Material Batch Tab Fields

The following table defines the fields used in the material_batches tab. The machine-readable definitions of these vocabulary entries are provided in material_batches_tab_fields.json

Field Name	Definition
mixing_protocol_file_name	File name associated with the documented material mixing procedure or batching protocol.
batch_label	Original batch designation or identifier used in the source documentation or laboratory records.
material_class	General material classification associated with the batch.
cement_type	Type or designation of cement used in the mixture.

Field Name	Definition
printable	Boolean field indicating whether the mixture was designed for additive manufacturing or extrusion-based placement processes.
self_consolidating	Boolean field indicating whether the mixture qualifies as self-consolidating concrete or mortar.
cement_content_kg_m3	Mass of cement per cubic meter of material.
water_binder_ratio	Ratio between total water content and total binder content by mass.
silica_fume_content_kg_m3	Mass of silica fume per cubic meter of material.
fly_ash_content_kg_m3	Mass of fly ash per cubic meter of material.
fly_ash_class	Classification or designation of fly ash used in the mixture.
slag_content_kg_m3	Mass of slag or ground granulated blast furnace slag per cubic meter of material.
fine_aggregate_content_kg_m3	Mass of fine aggregate per cubic meter of material.
fine_aggregate_dry_density_kg_m3	Dry density of the fine aggregate.
coarse_aggregate_content_kg_m3	Mass of coarse aggregate per cubic meter of material.
coarse_aggregate_dry_density_kg_m3	Dry density of the coarse aggregate.
max_aggregate_size_mm	Maximum nominal aggregate size.
aggregate_type	Type or mineralogical classification of aggregate used in the mixture.
superplasticizer_type	Type or commercial classification of superplasticizer admixture.
superplasticizer_content_kg_m3	Mass of superplasticizer admixture per cubic meter of material.
water_reducer_type	Type or classification of water-reducing admixture.
water_reducer_content_ml_m3	Volume of water-reducing admixture per cubic meter of material.
air_entrainment_type	Type or classification of air-entraining admixture.
air_entrainment_content_ml_m3	Volume of air-entraining admixture per cubic meter of material.
rheology_modifier_type	Type or classification of rheology-modifying admixture.
rheology_modifier_content_kg_m3	Mass of rheology-modifying admixture per cubic meter of material.
hydration_accelerator_type	Type or classification of hydration accelerator admixture.
hydration_accelerator_content_ml_m3	Volume of hydration accelerator admixture per cubic meter of material.
fiber_type	Type or material classification of fibers used in the mixture.
fiber_product_name	Commercial product name or designation of the fibers.
fiber_volume_fraction	Fiber volume fraction within the composite material.
fiber_length_mm	Nominal fiber length.
fiber_diameter_mm	Nominal fiber diameter.

Field Name	Definition
air_content_percent_volume_concrete	Measured air content expressed as percentage of total concrete volume.
date_of_pouring_yyyy-mm-dd	Date associated with specimen casting or material pouring using the ISO YYYY-MM-DD format.

9.3.1 Material Classes

The following table defines the controlled vocabulary for the material_class field used in the material_batches tab. The machine-readable definitions of these vocabulary entries are also provided in material_classes.json.

Term	Definition
cement_paste	A cementitious material composed primarily of cement, water, and optional chemical or mineral admixtures without fine or coarse aggregates.
engineered_cementitious_composites	A fiber-reinforced cementitious composite specifically engineered to exhibit strain-hardening behavior and distributed multiple cracking under tensile loading.
fiber_reinforced_concrete	Concrete containing discrete fibers intended to improve mechanical performance, toughness, crack control, impact resistance, or durability.
heavyweight_concrete	Concrete produced using heavyweight aggregates to achieve a density greater than that of conventional normal-weight concrete, typically for radiation shielding or specialized structural applications.
high_performance_concrete	Concrete designed to achieve enhanced mechanical, durability, rheological, or long-term performance characteristics beyond those of conventional concrete.
lightweight_concrete	Concrete produced using lightweight aggregates or other density-reducing techniques to achieve reduced self-weight compared to conventional concrete.
mortar	A cementitious composite consisting primarily of cement, water, fine aggregates, and optional admixtures, without coarse aggregates.
normal_weight_concrete	Conventional concrete produced using normal-density aggregates and having density characteristics typical of standard structural concrete.
ultra_high_performance_concrete	A highly optimized cementitious composite characterized by very high compressive strength, low permeability, enhanced durability, and typically containing fine powders and fibers.
other	A material class that does not fall within the predefined categories.

9.3.2 Cement Types

The following table defines the controlled vocabulary for the `cement_type` field used in the `material_batches` tab. The machine-readable definitions of these vocabulary entries are provided in `cement_types.json`.

Term	Definition
ASTM C1157 Hydraulic Cements	
`ASTM_C1157_Type_GU`	General use hydraulic cement conforming to ASTM C1157.
`ASTM_C1157_Type_HE`	High early-strength hydraulic cement conforming to ASTM C1157.
`ASTM_C1157_Type_HS`	High sulfate-resistance hydraulic cement conforming to ASTM C1157.
`ASTM_C1157_Type_LH`	Low-heat hydraulic cement conforming to ASTM C1157.
`ASTM_C1157_Type_MH`	Moderate-heat hydraulic cement conforming to ASTM C1157.
`ASTM_C1157_Type_MS`	Moderate sulfate-resistance hydraulic cement conforming to ASTM C1157.
ASTM C150 Portland Cements	
`ASTM_C150_Type_I`	General-purpose portland cement conforming to ASTM C150.
`ASTM_C150_Type_II`	Moderate sulfate-resistance portland cement conforming to ASTM C150.
`ASTM_C150_Type_III`	High early-strength portland cement conforming to ASTM C150.
`ASTM_C150_Type_IV`	Low-heat portland cement conforming to ASTM C150.
`ASTM_C150_Type_V`	High sulfate-resistance portland cement conforming to ASTM C150.
ASTM C595 Blended Hydraulic Cements	
`ASTM_C595_Type_IL`	Portland-limestone cement conforming to ASTM C595.
`ASTM_C595_Type_IL_HE`	High early-strength portland-limestone cement conforming to ASTM C595.
`ASTM_C595_Type_IL_HS`	High sulfate-resistance portland-limestone cement conforming to ASTM C595.
`ASTM_C595_Type_IL_MS`	Moderate sulfate-resistance portland-limestone cement conforming to ASTM C595.
`ASTM_C595_Type_IP`	Pozzolanic blended hydraulic cement conforming to ASTM C595.
`ASTM_C595_Type_IP_HE`	High early-strength pozzolanic blended hydraulic cement conforming to ASTM C595.
`ASTM_C595_Type_IP_HS`	High sulfate-resistance pozzolanic blended hydraulic cement conforming to ASTM C595.
`ASTM_C595_Type_IP_MS`	Moderate sulfate-resistance pozzolanic blended hydraulic cement conforming to ASTM C595.
`ASTM_C595_Type_IS_LS70_HS`	Slag blended hydraulic cement with high sulfate resistance conforming to ASTM C595.
`ASTM_C595_Type_IS_LT70`	Slag blended hydraulic cement conforming to ASTM C595.
`ASTM_C595_Type_IS_LT70_HE`	High early-strength slag blended hydraulic cement conforming to ASTM C595.
`ASTM_C595_Type_IS_LT70_MS`	Moderate sulfate-resistance slag blended hydraulic cement conforming to ASTM C595.

Term	Definition
ASTM_C595_Type_IT_S_LT70	Ternary blended hydraulic cement conforming to ASTM C595.
ASTM_C595_Type_IT_S_LT70_HE	High early-strength ternary blended hydraulic cement conforming to ASTM C595.
ASTM_C595_Type_IT_S_LT70_HS	High sulfate-resistance ternary blended hydraulic cement conforming to ASTM C595.
ASTM_C595_Type_IT_S_LT70_MS	Moderate sulfate-resistance ternary blended hydraulic cement conforming to ASTM C595.
EN 197 Common Cements	
EN197_CEM_I	Portland cement conforming to EN 197.
EN197_CEM_II	Portland-composite cement conforming to EN 197.
EN197_CEM_III	Blastfurnace cement conforming to EN 197.
EN197_CEM_IV	Pozzolanic cement conforming to EN 197.
EN197_CEM_V	Composite cement conforming to EN 197.
GB175 Chinese Cement Standards	
GB175_PC	Composite portland cement conforming to GB175.
GB175_PF	Fly ash portland cement conforming to GB175.
GB175_PI	Portland cement type I conforming to GB175.
GB175_PII	Portland cement type II conforming to GB175.
GB175_PO	Ordinary portland cement conforming to GB175.
GB175_PS	Slag portland cement conforming to GB175.
other	Cement type that does not fall within the predefined categories.
unknown	Cement type not reported or not identifiable from the available documentation.

9.3.3 Superplasticizer Types

The following table defines the controlled vocabulary for the superplasticizer_type field used in the material_batches tab. The machine-readable definitions of these vocabulary entries are provided insuperplasticizer_types.json.

Term	Definition
superplasticizer_unspecified	Superplasticizer or high-range water-reducing admixture reported without a more specific chemical classification.
melamine_sulfonate	High-range water-reducing admixture based on sulfonated melamine formaldehyde polymers.
naphthalene_sulfonate	High-range water-reducing admixture based on sulfonated naphthalene formaldehyde polymers.
polycarboxylate_ether	High-range water-reducing admixture based on polycarboxylate ether chemistry.
polycarboxylate_modified	Modified polycarboxylate-based admixture with proprietary or chemically altered formulations.
other	Superplasticizer type that does not fall within the predefined categories.
unknown	Superplasticizer type not reported or not identifiable from the available documentation.

9.3.4 Water Reducer Types

The following table defines the controlled vocabulary for the `water_reducer_type` field used in the `material_batches` tab. The machine-readable definitions of these vocabulary entries are provided in `water_reducer_types.json`.

Term	Definition
`water_reducer_unspecified`	Water-reducing admixture reported without a more specific chemical or performance classification with the exception of high-range water-reducing admixture, which is classified as superplasticizer.
`normal_range_water_reducer`	Normal-range water-reducing admixture used to reduce water demand while maintaining workability.
`mid_range_water_reducer`	Mid-range water-reducing admixture providing moderate water reduction and workability enhancement.
`set_retarding_water_reducer`	Water-reducing admixture formulated to delay cement hydration and setting time.
`set_accelerating_water_reducer`	Water-reducing admixture formulated to accelerate cement hydration and early-age strength development.
`lignosulfonate`	Water-reducing admixture based on lignosulfonate chemistry.
`hydroxycarboxylic_acid`	Water-reducing admixture based on hydroxycarboxylic acid chemistry.
`carbohydrate_based`	Water-reducing admixture based on carbohydrate-derived compounds or saccharide chemistry.
`other`	Water-reducer type that does not fall within the predefined categories.
`unknown`	Water-reducer type not reported or not identifiable from the available documentation.

9.3.5 Air-Entraining Admixture Types

The following table defines the controlled vocabulary for the `air_entrainment_type` field used in the `material_batches` tab. The machine-readable definitions of these vocabulary entries are provided in `air_entrainment_types.json`.

Term	Definition
`air_entrainment_unspecified`	Air-entraining admixture reported without a more specific chemical classification.
`synthetic_detergent_based`	Air-entraining admixture based on synthetic detergent or synthetic surfactant chemistry.
`vinsol_resin`	Air-entraining admixture based on vinsol resin compounds.
`wood_resin`	Air-entraining admixture derived from wood resin materials.
`fatty_acid_based`	Air-entraining admixture based on fatty acid compounds or derivatives.
`sulfated_hydrocarbon`	Air-entraining admixture based on sulfated hydrocarbon chemistry.
`protein_based`	Air-entraining admixture based on protein-derived compounds.

Term	Definition
natural_resin	Air-entraining admixture based on naturally occurring resin materials.
surfactant_based	Air-entraining admixture based on general surfactant chemistry without further specification.
other	Air-entraining admixture type that does not fall within the predefined categories.
unknown	Air-entraining admixture type not reported or not identifiable from the available documentation.

9.3.6 Rheology Modifier Types

The following table defines the controlled vocabulary for the rheology_modifier_type field used in the material_batches tab. The machine-readable definitions of these vocabulary entries are provided in rheology_modifier_types.json.

Term	Definition
rheology_modifier_unspecified	Rheology-modifying admixture reported without a more specific chemical or material classification.
viscosity_modifying_admixture	General viscosity-modifying admixture used to increase viscosity and improve segregation resistance.
cellulose_ether	Rheology-modifying admixture based on cellulose ether chemistry.
welan_gum	Biopolymer-based rheology modifier derived from welan gum polysaccharides.
xanthan_gum	Biopolymer-based rheology modifier derived from xanthan gum polysaccharides.
starch_based	Rheology-modifying admixture based on starch-derived compounds.
silica_based	Rheology-modifying admixture based on silica particles or silica-derived materials.
nanoclay_based	Rheology-modifying admixture based on nanoclay particles or nanostructured clay materials.
attapulgite_clay	Rheology-modifying admixture based on attapulgite clay minerals.
bentonite_based	Rheology-modifying admixture based on bentonite clay materials.
biopolymer_based	Rheology-modifying admixture based on naturally derived biopolymers.
synthetic_polymer_based	Rheology-modifying admixture based on synthetic polymer chemistry.
polyacrylamide_based	Rheology-modifying admixture based on polyacrylamide polymers.
diutan_gum	Biopolymer-based rheology modifier derived from diutan gum polysaccharides.
other	Rheology-modifier type that does not fall within the predefined categories.

Term	Definition
unknown	Rheology-modifier type not reported or not identifiable from the available documentation.

9.3.7 Hydration Accelerator Types

The following table defines the controlled vocabulary for the hydration_accelerator_type field used in the material_batches tab. The machine-readable definitions of these vocabulary entries are provided in hydration_accelerator_types.json.

Term	Definition
hydration_accelerator_unspecified	Hydration-accelerating admixture reported without a more specific chemical or performance classification.
calcium_chloride	Hydration accelerator based on calcium chloride chemistry commonly used to accelerate setting and early-age strength development.
non_chloride_accelerator	Hydration accelerator formulated without chloride compounds to minimize corrosion risks in reinforced concrete.
calcium_nitrate	Hydration accelerator based on calcium nitrate chemistry.
calcium_formate	Hydration accelerator based on calcium formate chemistry.
triethanolamine	Hydration accelerator based on triethanolamine compounds.
sodium_thiocyanate	Hydration accelerator based on sodium thiocyanate chemistry.
sodium_nitrate	Hydration accelerator based on sodium nitrate chemistry.
aluminate_based	Hydration accelerator based on aluminate-containing compounds.
silicate_based	Hydration accelerator based on silicate-containing compounds.
carbonate_based	Hydration accelerator based on carbonate-containing compounds.
alkali_based	Hydration accelerator based on alkali-containing compounds.
shotcrete_accelerator	Hydration accelerator specifically formulated for shotcrete applications requiring rapid stiffening and setting.
set_accelerating_admixture	Admixture primarily intended to reduce setting time and accelerate stiffening behavior.
hardening_accelerating_admixture	Admixture primarily intended to accelerate hydration and early-age hardening.
other	Hydration-accelerator type that does not fall within the predefined categories.

Term	Definition
unknown	Hydration-accelerator type not reported or not identifiable from the available documentation.

9.3.8 Aggregate Types

The following table defines the controlled vocabulary for the aggregate_type field used in the material_batches tab. The machine-readable definitions of these vocabulary entries are provided in aggregate_types.json .

Term	Definition
basalt	Aggregate derived from basaltic igneous rock.
calcareous	Aggregate primarily composed of calcium carbonate-based minerals.
dolomite	Aggregate derived from dolomitic rock formations.
granite	Aggregate derived from granitic igneous rock.
gravel_crushed	Crushed gravel aggregate produced through mechanical crushing processes.
gravel_natural	Naturally occurring gravel aggregate with minimal processing.
limestone	Aggregate derived from limestone rock formations.
lightweight_expanded_clay	Lightweight aggregate produced from expanded clay materials.
lightweight_pumice	Lightweight aggregate derived from pumice volcanic material.
lightweight_shale	Lightweight aggregate produced from expanded shale materials.
lightweight_slate	Lightweight aggregate produced from expanded slate materials.
manufactured_sand	Artificially manufactured fine aggregate produced through crushing operations.
natural_sand	Naturally occurring fine aggregate such as river or pit sand.
quartzite	Aggregate derived from quartzite metamorphic rock.
recycled_concrete	Recycled aggregate obtained from crushed concrete materials.
recycled_masonry	Recycled aggregate obtained from crushed masonry materials.
river_gravel	Naturally rounded gravel aggregate obtained from river deposits.
sandstone	Aggregate derived from sandstone rock formations.
siliceous	Aggregate primarily composed of silica-rich minerals.
slag	Aggregate produced from metallurgical slag byproducts.
other	Aggregate type that does not fall within the predefined categories.
unknown	Aggregate type not reported or not identifiable from the available documentation.

9.3.9 Fiber Types

The following table defines the controlled vocabulary for the fiber_type field used in the material_batches tab. The machine-readable definitions of these vocabulary entries are provided in fiber_types.json.

Term	Definition
aramid	Fiber based on aramid polymer chemistry characterized by high tensile strength and toughness.
basalt	Mineral fiber produced from melted basalt rock.
carbon	Fiber based on carbon filament materials characterized by high stiffness and strength.
glass	Fiber produced from glass materials commonly used for crack control and reinforcement.
natural	Fiber derived from naturally occurring biological or plant-based materials.
polyethylene	Synthetic polymer fiber based on polyethylene materials.
polypropylene_macro	Macro-scale polypropylene fiber typically used for structural crack control and toughness enhancement.
polypropylene_micro	Micro-scale polypropylene fiber typically used for plastic shrinkage cracking mitigation and spalling resistance.
polyvinyl_alcohol	Synthetic polymer fiber based on polyvinyl alcohol chemistry, commonly used in engineered cementitious composites.
recycled_polymer	Fiber produced from recycled polymeric materials.
steel_crimped	Steel fiber with crimped geometry used to enhance mechanical anchorage and pullout resistance.
steel_hooked	Steel fiber with hooked ends used to improve bond and pullout resistance.
steel_hybrid	Combination of different steel fiber geometries, dimensions, or material properties within the same mixture.
steel_straight	Straight steel fiber without mechanical end anchorage.
steel_twisted	Steel fiber with twisted geometry used to enhance mechanical interlock and pullout resistance.
other	Fiber type that does not fall within the predefined categories.
unknown	Fiber type not reported or not identifiable from the available documentation.

9.4 Geometries Tab Fields

The following table defines the fields used in the geometry tab. The machine-readable definitions of these vocabulary entries are provided in geometries_tab_fields.json

Field Name	Definition
specimen_geometry_type	Classification of the specimen geometry according to the standardized geometry parametrization library.
specimen_geometry_file_name	File name associated with the geometry parameterization file describing the specimen geometry dimensions, coordinate system, and geometric parameters.

9.4.1 Specimen Geometry Types

The following table defines the controlled vocabulary for the specimen_geometry_type field used in the geometries tab. The machine-readable definitions of these vocabulary entries are provided in specimen_geometries_types.json.

Term	Definition
cube	Cube-shaped specimen geometry.
prism	Rectangular prismatic specimen geometry.
cylinder	Cylindrical specimen geometry.
dogbone	Dogbone-shaped specimen geometry.
ell_shape	L-shaped specimen geometry.
notched_cylinder	Cylindrical specimen geometry containing a notch.
notched_prism	Prismatic specimen geometry containing a notch with a possible offset with respect to the centerline.
double_notched_prism	Prismatic specimen geometry containing a 2 notches located anti-symmetrically with respect to the centerline .
pci_double_tee_10ft	Prestressed concrete PCI double-tee section with nominal 10-foot flange width.
pci_double_tee_12ft	Prestressed concrete PCI double-tee section with nominal 12-foot flange width.
pci_double_tee_8ft	Prestressed concrete PCI double-tee section with nominal 8-foot flange width.
aashto_bulb_tee_BT54	AASHTO bulb-tee prestressed girder section BT-54.
aashto_bulb_tee_BT63	AASHTO bulb-tee prestressed girder section BT-63.
aashto_bulb_tee_BT72	AASHTO bulb-tee prestressed girder section BT-72.
aashto_girder_type_I	AASHTO prestressed girder section Type I.
aashto_girder_type_II	AASHTO prestressed girder section Type II.
aashto_girder_type_III	AASHTO prestressed girder section Type III.
aashto_girder_type_IV	AASHTO prestressed girder section Type IV.
aashto_girder_type_V	AASHTO prestressed girder section Type V.
aashto_girder_type_VI	AASHTO prestressed girder section Type VI.

9.5 Reinforcement Materials Tab Fields

The following table defines the fields used in the reinforcement_materials tab.

Field Name	Definition
GENERAL	
reinforcement_material_type	Material classification associated with the reinforcement system.
stress_strain_curve_file	File name associated with the reinforcement constitutive stress-strain relationship.
elastic_modulus_mpa	Elastic modulus of the reinforcement material expressed in MPa.
yield_strength_mpa	Yield strength of the reinforcement material expressed in MPa.
rupture_strength_mpa	Ultimate or rupture strength of the reinforcement material expressed in MPa.
rupture_strain	Strain corresponding to rupture or ultimate failure of the reinforcement material.

9.5.1 Reinforcement Material Types

The following table defines the controlled vocabulary for the reinforcement_material field used in the reinforcement tab. The machine-readable definitions of these vocabulary entries are provided in reinforcement_material_types.json.

Term	Definition
basalt_fiber_reinforced_polymer	Fiber-reinforced polymer reinforcement composed of basalt fibers embedded within a polymeric matrix.
black_steel	Conventional carbon steel reinforcement without corrosion-resistant coating.
carbon_fiber_reinforced_polymer	Fiber-reinforced polymer reinforcement composed of carbon fibers embedded within a polymeric matrix.
galvanized_steel	Steel reinforcement protected through galvanization or zinc coating processes.
glass_fiber_reinforced_polymer	Fiber-reinforced polymer reinforcement composed of glass fibers embedded within a polymeric matrix.
stainless_steel	Corrosion-resistant steel reinforcement based on stainless steel alloys.
other	Reinforcement material type that does not fall within the predefined categories.

9.6 Specimen Tab Fields

9.6.1 Structural Form Types

The following table defines the controlled vocabulary for the structural_form_type field used in the specimens tab. The machine-readable definitions of these vocabulary entries are provided in structural_form_types.json.

Term	Definition
arch	Structural form primarily resisting loads through compressive arching action.
beam	Structural form primarily resisting transverse loads through flexure and shear actions.
column	Structural form primarily resisting axial compressive loading.
slab	Plate-like structural form primarily resisting transverse loading through flexural plate action.
wall	Planar vertical structural form resisting axial, flexural, shear, or in-plane loading actions.
other	Structural form type that does not fall within the predefined categories.

9.7 Reinforcement Layout Tab Fields

The following table defines the fields used in the reinforcement_layouts tab. The machine-readable definitions of these vocabulary entries are provided in reinforcement_layouts_tab_fields.json

Field Name	Definition
longitudinal_reinforcement	Boolean field indicating whether longitudinal reinforcement is present.
transverse_reinforcement	Boolean field indicating whether transverse reinforcement is present.
external_reinforcement	Boolean field indicating whether external reinforcement is present.

Field Name	Definition
reinforcement_file_name	reinforcement_file_name: Name of an optional supplementary file containing drawings, CAD files, annotated figures, images, or other documentation describing the reinforcement configuration associated with the reinforcement component.

9.8 Reinforcement Components Tab Fields

The following table defines the fields used in the reinforcement_components tab. The machine-readable definitions of these vocabulary entries are provided in reinforcement_components_tab_fields.json

Field Name	Definition
GENERAL	
reinforcement_type	Classification of the reinforcement configuration according to the reinforcement parametrization library.
reinforcement_orientation	Primary orientation or alignment of the reinforcement relative to the specimen coordinate system.
FLEXURAL, TENSILE, STIRRUPS, TIES	
number	Number of reinforcement elements, bars, strands, or layers associated with the reinforcement configuration.
bar_designation	Nominal reinforcement designation or standard bar identification.
bar_nominal_diameter_mm	Nominal reinforcement diameter expressed in millimeters.
bar_nominal_area_mm2	Nominal reinforcement cross-sectional area expressed in square millimeters.
layer_depth_mm	Depth of the reinforcement layer relative to the reference geometry coordinate system.
spacing_mm	Center-to-center spacing between adjacent reinforcement elements expressed in millimeters.
horizontal_cover_mm	Horizontal concrete cover distance measured from the specimen surface to the reinforcement.
vertical_cover_mm	Vertical concrete cover distance measured from the specimen surface to the reinforcement.
begin_distance_mm	Distance from the geometry reference origin to the beginning of the reinforcement region.
end_distance_mm	Distance from the geometry reference origin to the end of the reinforcement region.
EXTERNAL REINFORCEMENT	
reinforcement_centroid_x_mm	Global x-coordinate of the centroid of the external reinforcement system.
reinforcement_centroid_y_mm	Global y-coordinate of the centroid of the external reinforcement system.
reinforcement_centroid_z_mm	Global z-coordinate of the centroid of the external reinforcement system.
reinforcement_width_mm	Width of the external reinforcement element expressed in millimeters.

Field Name	Definition
reinforcement_length_mm	Length of the external reinforcement element expressed in millimeters.
reinforcement_thickness_mm	Thickness of the external reinforcement element expressed in millimeters.
reinforcement_angle_deg	Orientation angle of the external reinforcement relative to the specimen coordinate system expressed in degrees.
adhesive_type	Classification of the adhesive material used to bond the external reinforcement system.
adhesive_thickness_mm	Thickness of the adhesive layer expressed in millimeters.
adhesive_modulus_mpa	Elastic modulus of the adhesive material expressed in MPa.
adhesive_strength_mpa	Strength of the adhesive material expressed in MPa.
adhesive_failure_strain	Failure strain associated with the adhesive material.
anchorage_presence	Boolean field indicating whether anchorage devices are present in the external reinforcement system.
anchorage_type	Classification of the anchorage system associated with the external reinforcement.
anchorage_size_mm	Characteristic size or dimension of the anchorage system expressed in millimeters.
anchorage_location_x_mm	Global x-coordinate of the anchorage location.
anchorage_location_y_mm	Global y-coordinate of the anchorage location.
anchorage_location_z_mm	Global z-coordinate of the anchorage location.
anchorage_modulus_mpa	Elastic modulus of the anchorage material expressed in MPa.
anchorage_strength_mpa	Strength of the anchorage material expressed in MPa.
anchorage_failure_strain	Failure strain associated with the anchorage material.

9.8.1 Reinforcement Types

The following table defines the controlled vocabulary for the reinf_type field used in the reinforcement tab. The machine-readable definitions of these vocabulary entries are provided in reinforcement_types.json.

Term	Definition
2_legs_stirrups	Transverse reinforcement configuration composed of two-legged stirrups typically used for shear reinforcement and confinement.
plates	Externally attached or bonded reinforcement system composed of plate-like reinforcement elements.
sheets	Externally bonded reinforcement system composed of thin sheet-like materials such as FRP sheets.
tensile_bar	Longitudinal reinforcing bar primarily intended to resist tensile stresses.
ties	Closed transverse reinforcement elements used for confinement and stability of longitudinal reinforcement.
wraps	Externally wrapped reinforcement systems surrounding portions of the specimen for confinement or strengthening purposes.
other	Reinforcement type that does not fall within the predefined categories.

9.8.2 Bar Designations

The following table defines the controlled vocabulary for the `bar_designation` field used in the `reinforcement` tab. The machine-readable definitions of these vocabulary entries are provided in `bar_designation_list.json` .

Term	Definition
ASTM Reinforcing Bar Designations	
ASTM_#3	ASTM reinforcing bar designation #3.
ASTM_#4	ASTM reinforcing bar designation #4.
ASTM_#5	ASTM reinforcing bar designation #5.
ASTM_#6	ASTM reinforcing bar designation #6.
ASTM_#7	ASTM reinforcing bar designation #7.
ASTM_#8	ASTM reinforcing bar designation #8.
ASTM_#9	ASTM reinforcing bar designation #9.
ASTM_#10	ASTM reinforcing bar designation #10.
ASTM_#11	ASTM reinforcing bar designation #11.
ASTM_#14	ASTM reinforcing bar designation #14.
ASTM_#18	ASTM reinforcing bar designation #18.
EN Reinforcing Bar Designations	
EN_Ø6	European reinforcing bar designation with nominal diameter 6 mm.
EN_Ø8	European reinforcing bar designation with nominal diameter 8 mm.
EN_Ø10	European reinforcing bar designation with nominal diameter 10 mm.
EN_Ø12	European reinforcing bar designation with nominal diameter 12 mm.
EN_Ø14	European reinforcing bar designation with nominal diameter 14 mm.
EN_Ø16	European reinforcing bar designation with nominal diameter 16 mm.
EN_Ø20	European reinforcing bar designation with nominal diameter 20 mm.
EN_Ø25	European reinforcing bar designation with nominal diameter 25 mm.
EN_Ø28	European reinforcing bar designation with nominal diameter 28 mm.
EN_Ø32	European reinforcing bar designation with nominal diameter 32 mm.
EN_Ø40	European reinforcing bar designation with nominal diameter 40 mm.
GB Reinforcing Bar Designations	
GB_6	Chinese reinforcing bar designation with nominal diameter 6 mm.
GB_8	Chinese reinforcing bar designation with nominal diameter 8 mm.
GB_10	Chinese reinforcing bar designation with nominal diameter 10 mm.
GB_12	Chinese reinforcing bar designation with nominal diameter 12 mm.
GB_14	Chinese reinforcing bar designation with nominal diameter 14 mm.
GB_16	Chinese reinforcing bar designation with nominal diameter 16 mm.
GB_18	Chinese reinforcing bar designation with nominal diameter 18 mm.
GB_20	Chinese reinforcing bar designation with nominal diameter 20 mm.
GB_22	Chinese reinforcing bar designation with nominal diameter 22 mm.
GB_25	Chinese reinforcing bar designation with nominal diameter 25 mm.
GB_28	Chinese reinforcing bar designation with nominal diameter 28 mm.
GB_32	Chinese reinforcing bar designation with nominal diameter 32 mm.
GB_36	Chinese reinforcing bar designation with nominal diameter 36 mm.
GB_40	Chinese reinforcing bar designation with nominal diameter 40 mm.
JIS Reinforcing Bar Designations	
JIS_D6	Japanese reinforcing bar designation D6.

Term	Definition
JIS_D10	Japanese reinforcing bar designation D10.
JIS_D13	Japanese reinforcing bar designation D13.
JIS_D16	Japanese reinforcing bar designation D16.
JIS_D19	Japanese reinforcing bar designation D19.
JIS_D22	Japanese reinforcing bar designation D22.
JIS_D25	Japanese reinforcing bar designation D25.
JIS_D29	Japanese reinforcing bar designation D29.
JIS_D32	Japanese reinforcing bar designation D32.
JIS_D35	Japanese reinforcing bar designation D35.
JIS_D38	Japanese reinforcing bar designation D38.
JIS_D41	Japanese reinforcing bar designation D41.
other	Bar designation that does not fall within the predefined categories.

9.8.3 Adhesive Types

The following table defines the controlled vocabulary for the adhesive_type field used in the reinforcement tab. The machine-readable definitions of these vocabulary entries are provided in adhesive_types.json.

Term	Definition
cementitious	Adhesive or bonding material based on cementitious mortar or grout formulations.
epoxy	Adhesive system based on epoxy resin chemistry commonly used for externally bonded reinforcement applications.
methacrylate	Adhesive system based on methacrylate resin chemistry.
polyester	Adhesive system based on polyester resin chemistry.
polyurethane	Adhesive system based on polyurethane polymer chemistry.
vinyl_ester	Adhesive system based on vinyl ester resin chemistry.
other	Adhesive type that does not fall within the predefined categories.

9.8.4 Anchorage Types

The following table defines the controlled vocabulary for the anchorage_type field used in the reinforcement tab. The machine-readable definitions of these vocabulary entries are provided in anchorage_types.json.

Term	Definition
adhesive_anchor	Anchorage system based primarily on adhesive bonding between the anchor and the surrounding substrate.
embedded_bar_anchor	Anchorage system using embedded reinforcement bars or rods to transfer forces into the substrate.
end_plate_anchor	Anchorage system using end plates to mechanically transfer and distribute reinforcement forces.
frp_fan_anchor	Fiber-reinforced polymer anchorage system using fan-shaped fiber bundles to improve force transfer and delay debonding.
frp_full_wrap	Anchorage system using continuous full wrapping of FRP reinforcement around the structural member.

Term	Definition
frp_patch_anchor	Fiber-reinforced polymer anchorage system using localized FRP patches for reinforcement anchorage.
frp_spike_anchor	Fiber-reinforced polymer anchorage system using spike-shaped embedded fiber bundles.
frp_u_wrap	Fiber-reinforced polymer anchorage system using U-shaped wrapping configurations.
mechanical_anchor_bolt	Mechanical anchorage system using bolts or threaded fasteners.
mechanical_anchor_plate	Mechanical anchorage system using bearing plates or clamping plates to transfer forces.
near_surface_mounted_anchor	Anchorage system associated with near-surface-mounted reinforcement configurations.
other	Anchorage type that does not fall within the predefined categories.

9.9 Test Fields

The following table defines the fields used in the tests tab. The machine-readable definitions of these vocabulary entries are provided in tests_tab_fields.json.

Field Name	Definition
test_id	Unique identifier associated with the experimental test within the database.
test_setup_visualization_file_name	File name associated with images, schematics, photographs, or visualizations describing the experimental test setup.
test_protocol_file_name	File name associated with the experimental test protocol, procedure, or testing standard documentation.
test_type	Classification of the experimental test according to the standardized test type vocabulary.
test_parameters_file_name	File name associated with the test parameterization file describing loading configurations, boundary conditions, geometric parameters, or other test-specific information.
main_observed_mechanisms	Primary physical, mechanical, chemical, thermal, transport, durability, or fracture mechanisms observed during the test.
age_days	Specimen age expressed in days at the time of testing.
curing_condition	Classification of the curing environment or curing procedure applied prior to testing.
hygral_boundary_condition	Hygral boundary condition describing moisture exchange characteristics during testing.
thermal_boundary_condition	Thermal boundary condition describing heat transfer characteristics during testing.
initial_env_temperature_C	Initial environmental temperature at the beginning of testing expressed in degrees Celsius.

Field Name	Definition
env_temperature_C_history_file_name	File name associated with the recorded environmental temperature history during testing.
initial_env_relative_humidity_percent	Initial environmental relative humidity at the beginning of testing expressed as percentage.
env_relative_humidity_history_file_name	File name associated with the recorded environmental relative humidity history during testing.
test_date_yyyy-mm-dd	Date associated with the experimental test using the ISO YYYY-MM-DD format.

9.9.1 Test Types

The following table defines the controlled vocabulary for the test_type field used in the tests tab. The machine-readable definitions of these vocabulary entries are provided in test_types.json vocabulary.

Term	Definition
Generic Experimental Test Types	
accelerated_carbonation_test	Test evaluating carbonation penetration under accelerated environmental exposure conditions.
axial_test	General axial loading test performed under uniaxial compression or tension conditions.
biaxial_compression_test	Test involving biaxial compressive stress states applied to a specimen.
blast_test	Experimental test evaluating structural or material response under blast loading conditions.
chloride_profile_grinding	Test involving grinding and chemical analysis to determine chloride penetration profiles.
concrete_pullout_bond_test	Test evaluating bond behavior between concrete and embedded reinforcement through pullout loading.
corrosion_damaged_beam_test	Structural beam test performed on specimens affected by reinforcement corrosion.
creep_axial_compression_test	Long-term axial compression test evaluating creep deformation behavior.
creep_bending_test	Long-term flexural loading test evaluating creep behavior under bending.
cyclic_lateral_test	Test involving cyclic lateral loading typically used for structural or seismic investigations.
direct_tensile_test	Test evaluating tensile response under direct uniaxial tension.
double_punch_test	Compression-based punching test commonly used to estimate tensile or fracture properties.
drop_weight_impact_test	Impact test performed using a falling mass or drop-weight loading configuration.
eccentric_axial_compression_test	Axial compression test performed with eccentric loading producing combined compression and bending.
electrical_resistivity_test	Test measuring the electrical resistivity of cementitious materials or concrete.

Term	Definition
expansion_measurement	Measurement procedure evaluating expansion due to chemical, thermal, or durability-related mechanisms.
fiber_pullout_test	Test evaluating pullout resistance and bond behavior of fibers embedded in a matrix.
flow_through_leaching_test	Test evaluating material degradation under flowing leaching conditions.
four_point_bending_test	Flexural test performed using two loading points producing a constant moment region.
fully_adiabatic_calorimetry_test	Calorimetry test performed under fully adiabatic thermal conditions.
gas_permeability_test	Test measuring gas permeability of porous materials.
gap_test	Experimental test involving controlled gap geometry or discontinuity conditions.
hammer_test	Impact or rebound-based test performed using a hammer device.
heat_evolution_test	Test evaluating heat generation during hydration or chemical reactions.
heat_of_hydration_test	Test measuring the heat released during cement hydration.
hydrostatic_test	Test involving hydrostatic pressure loading conditions.
ICP_analysis_test	Chemical analysis performed using inductively coupled plasma techniques.
immersion_in_acid_test	Durability test involving immersion in acidic environments.
immersion_in_deionized_water_test	Durability or transport test involving immersion in deionized water.
interface_shear_transfer_test	Test evaluating shear transfer across material interfaces or construction joints.
length_change_test	Test evaluating shrinkage, swelling, or thermal length change behavior.
linear_polarization_resistance_LPR	Electrochemical test evaluating corrosion activity using linear polarization resistance methods.
mercury_intrusion_porosimetry_test	Test characterizing pore-size distribution using mercury intrusion porosimetry.
mini_slump_test	Reduced-scale slump-flow test commonly used for paste or mortar characterization.
modulus_of_rupture_test	Flexural test used to determine modulus of rupture or flexural tensile strength.
noru_mohammed_test	Specialized experimental test following the Noru-Mohammed procedure and measure mixed-mode fracture properties.
nuclear_magnetic_resonance_test	Test using nuclear magnetic resonance techniques for material characterization.
punching_shear_test	Structural test evaluating punching shear resistance.
rate_of_absorption_of_water_test	Test measuring capillary absorption or water uptake rate.

Term	Definition
reactivity_of_scm_test	Test evaluating reactivity of supplementary cementitious materials.
rebound_number_test	Surface hardness test based on rebound hammer measurements.
rotational_rheometry_test	Rheological characterization using rotational rheometry equipment.
scanning_electron_microscopy_test	Microscopy-based material characterization using scanning electron microscopy.
scaling_resistance_test	Durability test evaluating resistance to surface scaling.
semi_adiabatic_calorimetry_test	Calorimetry test performed under semi-adiabatic thermal conditions.
shear_friction_test	Test evaluating shear friction resistance across interfaces or cracks.
slump_flow_test	Workability test evaluating slump flow spread.
slump_test	Workability test measuring conventional slump.
splitting_tensile_test	Indirect tensile test performed by diametral compression loading.
surface_resistivity_test	Test measuring surface electrical resistivity.
tension_chord_test	Structural test evaluating tension stiffening or tension chord behavior.
thermal_diffusivity_test	Test measuring thermal diffusivity properties.
thermogravimetric_analysis_test	Material characterization using thermogravimetric analysis.
three_point_bending_test	Flexural test performed using a single central loading point.
time_of_setting_test	Test evaluating initial and final setting times.
torsion_compression_test	Combined torsion and compression loading test.
torsion_test	Test evaluating response under torsional loading.
triaxial_test	Test performed under triaxial stress conditions.
true_triaxial_test	Test performed under independently controlled three-dimensional stress states.
ultrasonic_pulse_velocity_test	Non-destructive test measuring ultrasonic pulse velocity.
uniaxial_strain_test	Test performed under constrained uniaxial strain conditions.
v_funnel_test	Workability test evaluating flow time through a V-funnel apparatus.
vane_shear_test	Test measuring shear resistance using vane shear equipment.
water_penetration_test	Durability test evaluating water penetration resistance.
wedge_splitting_test	Fracture mechanics test performed using wedge-induced splitting loading.
XRD_Rietveld_analysis_test	Material characterization using X-ray diffraction and Rietveld analysis methods.

Term	Definition
Standardized Test Procedures	
AASHTO_T325	AASHTO standard procedure for estimating concrete strength using maturity methods.
AASHTO_T336	AASHTO standard test method for coefficient of thermal expansion of hydraulic cement concrete.
AC434	Acceptance criteria for masonry and concrete strengthening using FRCM and SRG composite systems.
ASTM_A944/A944M	ASTM standard test method for comparing bond strength of steel reinforcing bars to concrete.
ASTM_C1074	ASTM standard practice for estimating concrete strength by the maturity method.
ASTM_C1152	ASTM standard test method for acid-soluble chloride in mortar and concrete.
ASTM_C1202	ASTM rapid chloride permeability test procedure.
ASTM_C1218	ASTM standard test method for water-soluble chloride in mortar and concrete.
ASTM_C1260	ASTM accelerated mortar-bar test for alkali-silica reactivity potential.
ASTM_C1293	ASTM concrete prism test for alkali-silica reaction evaluation.
ASTM_C138/C138M	ASTM test method for density, yield, and air content of concrete.
ASTM_C143/C143M	ASTM standard slump test for hydraulic-cement concrete.
ASTM_C1556	ASTM standard test method for apparent chloride diffusion coefficient.
ASTM_C157/C157M	ASTM test method for length change of hardened hydraulic-cement mortar and concrete.
ASTM_C1585	ASTM test method for measurement of rate of absorption of water by hydraulic-cement concretes.
ASTM_C1609/C1609M	ASTM flexural performance test for fiber-reinforced concrete using beam specimens.
ASTM_C1611/C1611M	ASTM slump-flow test for self-consolidating concrete.
ASTM_C1679	ASTM test method for measuring heat evolution of hydraulic cementitious materials.
ASTM_C1702	ASTM standard test method for heat of hydration.
ASTM_C173/C173M	ASTM volumetric method for air content of freshly mixed concrete.
ASTM_C1876	ASTM test method for bulk electrical resistivity of cementitious materials.
ASTM_C1897	ASTM test method for reactivity of supplementary cementitious materials.
ASTM_C191	ASTM Vicat needle test for time of setting of hydraulic cement.
ASTM_C231/C231M	ASTM pressure method for air content of freshly mixed concrete.

Term	Definition
ASTM_C293/C293M	ASTM three-point bending flexural strength test.
ASTM_C311	ASTM practice for sampling and testing fly ash or natural pozzolans.
ASTM_C39/C39M	ASTM compressive strength test for cylindrical concrete specimens.
ASTM_C403/C403M	ASTM penetration resistance test for time of setting.
ASTM_C469/C469M	ASTM test method for static modulus of elasticity and Poisson's ratio.
ASTM_C496/C496M	ASTM splitting tensile strength test for cylindrical concrete specimens.
ASTM_C512/C512M	ASTM creep test under sustained compressive loading.
ASTM_C531	ASTM test method for linear shrinkage and coefficient of thermal expansion.
ASTM_C597	ASTM ultrasonic pulse velocity test method.
ASTM_C642	ASTM test method for density, absorption, and voids in hardened concrete.
ASTM_C666/C666M	ASTM resistance of concrete to rapid freezing and thawing.
ASTM_C672/C672M	ASTM scaling resistance test for concrete surfaces exposed to deicing chemicals.
ASTM_C78/C78M	ASTM third-point loading flexural strength test.
ASTM_C7957/D7957M	ASTM specification related to FRP reinforcing bars.
ASTM_C805/C805M	ASTM rebound hammer test method.
ASTM_D7958/D7958M	ASTM bonded FRP beam test method.
ASTM_D8337/D8337M	ASTM single-lap shear test for FRP bonded systems.
EN_12350-2	EN slump test procedure for fresh concrete.
EN_12350-6	EN density test for fresh concrete.
EN_12350-7	EN air-content test procedure for fresh concrete.
EN_12350-8	EN slump-flow test for self-compacting concrete.
EN_12350-9	EN V-funnel test for self-compacting concrete.
EN_12390-13	EN test method for secant modulus of elasticity.
EN_12390-16	EN shrinkage test procedure for hardened concrete.
EN_12390-17	EN creep test procedure for hardened concrete.
EN_12390-3	EN compressive strength test for hardened concrete specimens.
EN_12390-6	EN splitting tensile strength test.
EN_12390-8	EN water penetration under pressure test.
EN_12504-2	EN rebound hammer test for concrete.
EN_12504-4	EN ultrasonic pulse velocity test for concrete.
EN_14630	EN test method for carbonation depth in hardened concrete.
EN_14651	EN flexural tensile test for fiber-reinforced concrete.
NT_Build_492	NT Build migration test for chloride penetration resistance.
RILEM_RC6	RILEM pull-out bond test for reinforcement-concrete interaction.

Term	Definition
RILEM_TC_162-TDF	RILEM recommendations for testing steel fiber reinforced concrete.
RILEM_TC_50-FMC	RILEM three-point bending fracture test procedure.
USACE_CRD-C124	USACE procedure for determining specific heat of concrete materials.
USACE_CRD-C36	USACE test procedure for thermal diffusivity measurements.
USACE_CRD-C39-81	USACE test procedure for coefficient of thermal expansion measurements.

9.9.2 Observed Mechanisms

The following table defines the controlled vocabulary for the observed_mechanisms field used in the tests tab. The machine-readable definitions of these vocabulary entries are provided in observed_mechanisms.json.

Term	Definition
Bond Mechanisms	
bond | bond_slip_and_debonding	Slip, debonding, or interface failure occurring between bonded materials or reinforcement systems.
bond | steel_concrete_interaction	Mechanical interaction and load transfer mechanisms between steel reinforcement and concrete.
Chemical Mechanisms	
chemical | degree_of_reaction	Extent of hydration, pozzolanic reaction, or chemical conversion within the material system.
chemical | ion_concentration	Measurement or evolution of ionic species concentration within the material or pore solution.
chemical | mass_loss	Mass reduction associated with chemical degradation, dissolution, or decomposition.
chemical | material_reactivity	Chemical reactivity of constituent materials or supplementary cementitious materials.
chemical | microstructure_morphology	Microstructural morphology and phase arrangement observed through microscopy or imaging techniques.
chemical | phase_assemblage	Formation, evolution, or quantification of mineralogical or chemical phases.
chemical | pore_structure	Characterization or evolution of pore-size distribution and pore connectivity.
Durability Mechanisms	
durability | alkali_silica_reaction	Expansion and deterioration caused by alkali-silica reaction mechanisms.
durability | carbonation	Carbonation processes occurring within cementitious materials.
durability | carbonation_depth	Measurement or evolution of carbonation penetration depth.
durability | chloride_content	Measurement or evolution of chloride concentration within the material.

Term	Definition
durability | chloride_penetration	Transport and diffusion of chloride ions through porous materials.
durability | corrosion_current	Electrochemical corrosion current associated with reinforcement corrosion processes.
durability | corrosion_probability	Estimated or measured probability of corrosion initiation or propagation.
durability | freeze_thaw_damage	Damage caused by cyclic freezing and thawing exposure.
durability | length_change_expansion	Expansion or dimensional change associated with durability-related mechanisms.
durability | mass_loss	Mass reduction caused by degradation, leaching, corrosion, or chemical attack.
durability | permeability	Fluid transport and permeability characteristics relevant to durability performance.
durability | pore_structure_evolution	Evolution of porosity and pore structure during degradation or aging.
durability | sulfate_attack_expansion	Expansion and deterioration caused by sulfate attack mechanisms.
Elastic Mechanisms	
elastic | linear_elasticity	Linear elastic material response under loading.
Fiber Mechanisms	
fiber | fiber_bridging_and_pull_out_resistance	Fiber bridging and pullout mechanisms contributing to post-cracking resistance.
fiber | post_cracking_residual_strength	Residual load-carrying capacity after matrix cracking due to fiber reinforcement.
Fracture Mechanisms	
fracture | fracture_energy_dissipation	Energy dissipation associated with crack initiation and propagation.
fracture | fracture_propagation	Initiation and propagation of fractures or cracks within the material.
fracture | mixed_mode_dilatancy	Dilatancy effects occurring under mixed-mode fracture conditions.
fracture | mode_III	Fracture behavior associated with tearing or anti-plane shear loading.
fracture | parallel_stress_effect	Influence of parallel stress components on fracture response.
Fresh-State Mechanisms	
fresh | air_entrainment	Air entrainment behavior and formation of void systems in fresh materials.
fresh | density_and_consistency	Fresh-state density and volumetric consistency characteristics.
fresh | rheology_evolution	Time-dependent evolution of rheological properties in fresh materials.
fresh | segregation_and_stability	Material segregation resistance and fresh-state stability.

Term	Definition
`fresh | setting`	Initial and final setting behavior of cementitious materials.
`fresh | setting_and_early_age_stiffening`	Setting and early-age stiffness development.
`fresh | spread_diameter`	Spread diameter or flow spread measurements in fresh-state tests.
`fresh | workability`	Workability and flow characteristics of fresh materials.
`fresh | rheology`	Yield stress and plastic viscosity governing rheological response.
FRP Mechanisms	
`frp | bond`	Bond behavior and load transfer mechanisms involving FRP reinforcement systems.
Non-Destructive Testing Mechanisms	
`ndt | surface_hardness_rebound_response`	Surface hardness response measured through rebound-based non-destructive testing.
`ndt | wave_propagation_velocity`	Wave propagation characteristics measured through ultrasonic or dynamic testing.
Stiffness Mechanisms	
`stiffness | bending_stiffness`	Flexural stiffness response under bending loading.
`stiffness | crack_width`	Crack opening width development and evolution.
`stiffness | dynamic_response`	Dynamic structural or material response characteristics.
`stiffness | natural_frequency`	Natural frequency response of structural or material systems.
`stiffness | tension_stiffening`	Tension stiffening effects due to reinforcement-concrete interaction.
`stiffness | time_dependent_deflection`	Time-dependent deflection behavior under sustained loading.
`stiffness | time_dependent_strain`	Time-dependent strain evolution under sustained loading or environmental exposure.
Strength Mechanisms	
`strength | aggregate_interlock`	Load transfer through aggregate interlock mechanisms across cracks or interfaces.
`strength | axial_capacity`	Axial load-carrying capacity of the specimen or structural member.
`strength | bending_capacity`	Flexural or bending load-carrying capacity.
`strength | biaxial_failure`	Failure under biaxial stress conditions.
`strength | bond_slip`	Bond-slip behavior between reinforcement and surrounding material.
`strength | cohesion_adhesion`	Cohesive and adhesive strength mechanisms within or between materials.
`strength | compressive_strain_hardening`	Compressive strain-hardening material response.
`strength | concrete_cone_breakout`	Concrete cone breakout failure associated with anchorage or pullout systems.
`strength | concrete_cone_failure`	Concrete cone-shaped failure mechanism under localized loading.

Term	Definition
strength | creep_buckling	Buckling instability associated with creep deformation.
strength | dowel_action	Load transfer through dowel action of embedded reinforcement.
strength | ductility	Ability to sustain inelastic deformation prior to failure.
strength | energy_dissipation	Energy dissipation capacity under cyclic, impact, or fracture loading.
strength | failure_mode	Observed structural or material failure mode.
strength | impact_resistance	Resistance to impact or impulsive loading.
strength | shear_fracture	Localized tensile-shear fracture mechanisms.
strength | multiaxial_failure	Failure under combined multiaxial stress states.
strength | pozzolan_reactivity	Reactivity and contribution of pozzolanic materials to strength development.
strength | seismic_response	Structural response under seismic or cyclic loading conditions.
strength | shear_friction	Shear transfer resistance through frictional mechanisms.
strength | slag_reactivity	Reactivity and contribution of slag materials to mechanical performance.
strength | slab_column_connection_failure	Failure of slab-column connection regions under structural loading.
strength | crushing_failure	Combined tensile and compressive crushing failure mechanisms.
strength | tensile_failure	Failure governed primarily by tensile cracking or rupture.
strength | tensile_strength	Tensile strength capacity of the material or structural component.
strength | torsion_capacity	Load-carrying capacity under torsional loading.
strength | unconfined_compressive_failure	Compressive failure occurring without lateral confinement.
strength | vertex_effect	Stress concentration or failure effects associated with geometric vertices or corners.
Thermal Mechanisms	
thermal | adiabatic_temperature_rise	Temperature increase under adiabatic hydration or thermal conditions.
thermal | heat_capacity	Heat storage capacity and thermal inertia properties.
thermal | heat_evolution	Heat generation associated with hydration or chemical reactions.
thermal | linear_thermal_expansion	Linear dimensional change caused by temperature variation.
thermal | thermal_diffusivity	Thermal diffusivity governing transient heat transport.
Time-Dependent Mechanisms	
time_dependent | autogenous_shrinkage	Autogenous shrinkage associated with self-desiccation and hydration.

Term	Definition
time_dependent | creep	Time-dependent deformation under sustained loading.
time_dependent | drying_shrinkage	Shrinkage caused by moisture loss to the environment.
time_dependent | maturity	Concrete maturity evolution associated with time-temperature history.
Transport Mechanisms	
transport | absorption_and_porosity	Absorption characteristics and porosity-related transport behavior.
transport | chloride_diffusion	Transport and diffusion of chloride ions through porous materials.
transport | electrical_resistivity	Electrical resistivity associated with ionic transport and pore connectivity.
transport | gas_permeability	Gas permeability and gas transport properties.
transport | sorptivity	Capillary suction and sorptivity behavior.
transport | water_permeability	Water permeability and hydraulic transport properties.
Other	
other	Observed mechanism that does not fall within the predefined categories.

9.9.3 Curing Conditions

The following table defines the controlled vocabulary for the curing_condition field used in the tests tab. The machine-readable definitions of these vocabulary entries are provided in curing_conditions.json.

The vocabulary includes common curing environments and curing procedures used for concrete and cementitious materials prior to testing.

Term	Definition
air_curing	Curing condition in which the specimen is exposed to ambient air without controlled moisture supply.
autoclave_curing	Curing condition involving elevated temperature and pressure within an autoclave environment.
controlled_environment	Curing condition performed under controlled environmental temperature and humidity conditions.
field_curing	Curing condition representative of in-situ or field exposure conditions.
fog_room	Curing condition performed in a fog room with high humidity maintained through mist or vapor systems.
heat_curing	Curing condition involving elevated temperature exposure to accelerate hydration and strength development.
laboratory_air	Curing condition in standard laboratory ambient air conditions without intentional moisture control.
lime_saturated_water	Curing condition involving immersion in lime-saturated water.
membrane_curing	Curing condition using membrane-forming curing compounds to reduce moisture loss.

Term	Definition
no_curing	Condition in which no intentional curing procedure was applied.
sealed_curing	Curing condition in which the specimen is sealed to minimize moisture exchange with the environment.
standard_moist_curing	Standard moist curing condition performed according to established laboratory curing procedures or testing standards.
steam_curing	Curing condition involving exposure to steam to accelerate hydration and early-age strength development.
water_curing	Curing condition involving immersion or continuous exposure to water.
other	Curing condition that does not fall within the predefined categories.
unknown	Curing condition not reported or not identifiable from the available documentation.

9.9.4 Hygral and Thermal Boundary Conditions

The following table defines the controlled vocabulary for the hygral_boundary_condition and thermal_boundary_condition fields. The machine-readable definitions of these vocabulary entries are provided in hygro_thermal_boundary_conditions.json.

Term	Definition
sealed	Boundary condition preventing moisture exchange with the surrounding environment.
unsealed	Boundary condition allowing moisture exchange with the surrounding environment.
insulated	Boundary condition designed to prevent thermal transfer through the boundary surface.
uninsulated	Boundary condition allowing thermal transfer through the boundary surface.
unknown	Boundary condition not reported or not identifiable from the available documentation.

9.10 Devices and Supports Tab Fields

The following table defines the fields used in the devices_and_supports tab. The machine-readable definitions of these vocabulary entries are provided in devices_and_supports_tab_fields.json.

Field Name	Definition
device_id	Unique identifier associated with the experimental device, sensor, fixture, or support system within the database.
device_type	Classification of the device, instrumentation system, loading fixture, or support according to the standardized devices-and-supports vocabulary.
x_location_mm	Global x-coordinate of the device or support location expressed in millimeters relative to the specimen coordinate system.
y_location_mm	Global y-coordinate of the device or support location expressed in millimeters relative to the specimen coordinate system.

Field Name	Definition
z_location_mm	Global z-coordinate of the device or support location expressed in millimeters relative to the specimen coordinate system.
orientation	Orientation or directional alignment of the device, sensor, support, or loading fixture relative to the specimen coordinate system.
device_parameters_file	File name associated with the device parameterization file describing geometry, mechanical properties, sensor specifications, support conditions, calibration information, or other device-specific parameters.

9.10.1 Device and Support Types

The following table defines the controlled vocabulary for the device_type field used in the devices_and_supports tab. The machine-readable definitions of these vocabulary entries are provided in devices_and_supports.json.

Term	Definition
accelerometer	Sensor used to measure acceleration and dynamic response.
adiabatic_calorimeter	Calorimetry device used to measure heat evolution under adiabatic conditions.
balance	Device used to measure mass or weight.
camera	General imaging device used for visual recording or monitoring.
clip_gauge	Displacement gauge typically attached across a crack or gauge length.
CMOD_gauge	Gauge used to measure crack mouth opening displacement.
CTOD_gauge	Gauge used to measure crack tip opening displacement.
data_logger	Electronic device used for automated acquisition and storage of sensor data.
DIC	Digital image correlation system used for full-field displacement and strain measurements.
diffusion_cell	Experimental apparatus used to evaluate diffusion and transport properties.
dilatometer	Instrument used to measure dimensional change or thermal expansion.
environmental_chamber	Controlled chamber used to regulate temperature, humidity, or environmental exposure conditions.
exposure_chamber	Chamber used to expose specimens to controlled environmental or chemical conditions.
extensometer	Instrument used to measure deformation or strain over a specified gauge length.
fixed_support	Rigid support restraining translational and rotational movement.
flow_table	Device used to evaluate flow characteristics and workability of fresh materials.
ftir_spectrometer	Fourier-transform infrared spectroscopy instrument used for chemical characterization.
half_cell_electrode	Electrochemical probe used to assess corrosion potential.

Term	Definition
high_speed_camera	High-frame-rate imaging device used to capture rapid events or fracture processes.
humidity_sensor	Sensor used to measure relative humidity.
icp	Inductively coupled plasma instrument used for chemical analysis.
isothermal_calorimeter	Calorimetry device used to measure heat evolution under isothermal conditions.
laser_displacement_sensor	Laser-based sensor used to measure displacement without physical contact.
laser_sensor	General laser-based sensing device for distance or position measurements.
leaching_reactor	Experimental reactor used to study leaching processes and chemical degradation.
length_comparator	Instrument used to measure specimen length changes with high precision.
load_cell	Force transducer used to measure applied load.
LVDT	Linear variable differential transformer used to measure displacement.
mercury_intrusion_porosimeter	Instrument used to characterize pore-size distributions through mercury intrusion.
micro_ct_scanner	Micro-computed tomography scanner used for three-dimensional internal imaging.
microscope	General optical or digital microscope used for imaging and material characterization.
migration_cell	Apparatus used to evaluate ionic migration and transport properties.
nmr_spectrometer	Nuclear magnetic resonance instrument used for material characterization.
optical_microscope	Optical microscope used for surface or microstructural observation.
pad_support	Compliant support element used to provide deformable boundary conditions.
penetration_resistance_device	Device used to evaluate penetration resistance and setting behavior.
permeability_apparatus	Experimental setup used to measure permeability properties.
ph_meter	Instrument used to measure pH values.
pin_support	Support allowing rotational movement while restraining translation.
potentiostat	Electrochemical instrument used for corrosion and electrochemical testing.
pressure_sensor	Sensor used to measure pressure.
rcpt_apparatus	Experimental apparatus used for rapid chloride permeability testing.
resistivity_meter	Instrument used to measure electrical resistivity.
rheometer	Instrument used to characterize rheological properties of fresh materials.

Term	Definition
roller_support	Support allowing translational movement in one direction while restraining other motions.
rotation_sensor	Sensor used to measure angular displacement or rotation.
scanning_electron_microscope	Electron microscope used for high-resolution microstructural characterization.
slump_cone	Conical apparatus used to perform slump tests on fresh concrete.
sorptivity_setup	Experimental setup used to evaluate capillary absorption and sorptivity.
strain_gauge	Sensor used to measure local strain.
string_potentiometer	Displacement transducer based on cable extension measurements.
temperature_sensor	Sensor used to measure temperature.
thermocouple	Temperature measurement device based on thermoelectric junctions.
tga_analyzer	Thermogravimetric analysis instrument used to evaluate mass changes with temperature.
unknown	Device type not reported or not identifiable from the available documentation.
vicat_apparatus	Apparatus used to determine setting times of cementitious materials.
wenner_probe	Probe configuration used to measure electrical resistivity using the Wenner method.
xrd_diffractometer	X-ray diffraction instrument used for phase identification and crystallographic characterization.
other	Device or support type that does not fall within the predefined categories.

9.11 Loading History Tab Fields

The following table defines the fields used in the loading_histories tab. The machine-readable definitions of these vocabulary entries are provided in loading_histories_tab_fields.json.

Field Name	Definition
loading_id	Unique identifier associated with the loading history within the database.
loading_mode	Classification of the loading regime according to the standardized loading mode vocabulary.
control_variable	Primary variable used to control the loading process during the experiment, such as load, displacement, strain, temperature, pressure, or other prescribed quantities.
initial_control_variable_rate	Initial rate associated with the applied control variable at the beginning of the loading history.
control_variable_rate_units	Units associated with the control-variable rate.
begin_time_sec	Initial time associated with the loading interval expressed in seconds.

Field Name	Definition
end_time_sec	Final time associated with the loading interval expressed in seconds.
loading_history_file_name	File name associated with the loading history record containing time-dependent loading information, control-variable evolution, or prescribed loading paths.

9.11.1 Loading Modes

The following table defines the controlled vocabulary for the loading_mode field used in the loading_histories tab. The machine-readable definitions of these vocabulary entries are provided in loading_modes.json.

Term	Definition
cyclic	Loading mode involving repeated loading and unloading cycles without systematic load reversal.
dynamic	Loading mode characterized by significant inertia or rate-dependent effects.
impulsive	Loading mode involving short-duration high-intensity loading pulses or impact-type actions.
monotonic	Loading mode characterized by continuously increasing or decreasing loading without unloading cycles.
random	Loading mode involving stochastic, irregular, or non-deterministic loading variations.
reversed_cyclic	Cyclic loading mode involving repeated load reversals between opposite loading directions or signs.
other	Loading mode that does not fall within the predefined categories.
unknown	Loading mode not reported or not identifiable from the available documentation.

9.11.2 Loading Control Variables

The following table defines the controlled vocabulary for the control_variable field used in the loading_histories tab. The machine-readable definitions of these vocabulary entries are provided in loading_control_variables.json .

Term	Definition
CMOD	Crack mouth opening displacement used as the primary loading control variable.
displacement	Prescribed displacement used as the primary loading control variable.
load	Applied force or load used as the primary loading control variable.
strain	Prescribed strain used as the primary loading control variable.
stress	Applied stress used as the primary loading control variable.
other	Control variable that does not fall within the predefined categories.
unknown	Control variable not reported or not identifiable from the available documentation.

9.12 Data Tab Fields

The following table defines the fields used in the `data` tab. The machine-readable definitions of these vocabulary entries are provided in `data_tab_fields.json`.

Field Name	Definition
GENERAL	
`data_id`	Unique identifier associated with the data entry within the database.
`notes`	Additional comments, annotations, assumptions, observations, or clarifications associated with the data entry.
`source_location`	Location within the original source document where the data were obtained, such as page numbers, figure identifiers, table identifiers, appendix references, or supplementary material locations.
`extraction_method`	Method used to extract, digitize, compute, or curate the reported data.
`extraction_date_yyyy-mm-dd`	Date associated with the data extraction process using the ISO YYYY-MM-DD format.
`curator_first_name`	First name of the curator responsible for data extraction or curation.
`curator_last_name`	Last name of the curator responsible for data extraction or curation.
`data_type`	Classification of the data according to the standardized data-type vocabulary.
`number_of_specimens`	Number of specimens associated with the reported data entry.
`specimen_label`	Original specimen designation or specimen label used in the source documentation.
SCALAR	
`device_id`	Identifier associated with the experimental device or instrumentation system used to obtain the reported quantity.
`quantity_reported`	Name or description of the experimentally reported quantity.
`quantity_reported_type`	Classification indicating whether the reported quantity is raw, derived, or otherwise categorized according to the corresponding vocabulary.
`quantity_reported_mean`	Mean or representative scalar value associated with the reported quantity.
`quantity_reported_standard_deviation`	Standard deviation associated with the reported scalar quantity.
`units`	Units associated with the reported quantity.
CURVE	
`x_device_id`	Identifier associated with the device measuring or controlling the x-axis quantity of the curve.

Field Name	Definition
y_device_id	Identifier associated with the device measuring or controlling the y-axis quantity of the curve.
x_quantity_reported	Name or description of the quantity represented along the x-axis of the curve.
x_quantity_reported_type	Classification associated with the x-axis quantity according to the quantity-reported-type vocabulary.
y_quantity_reported	Name or description of the quantity represented along the y-axis of the curve.
y_quantity_reported_type	Classification associated with the y-axis quantity according to the quantity-reported-type vocabulary.
file_name	File name associated with the curve dataset, numerical record, or external data file.
TABLE	
table_name	Name or title associated with the reported table dataset.
table_description	Description of the contents, organization, or purpose of the table dataset.
table_file_name	File name associated with the tabular dataset.
IMAGE / VIDEO	
name	Name or title associated with the image or video file.
description	Description of the image or video contents.
type	Classification of the image or video according to the corresponding media-type vocabulary.
format	File format associated with the image or video file.
filename	File name associated with the image or video asset.

9.12.1 Extraction Methods

The following table defines the controlled vocabulary for the extraction_method field used in the data tab. The machine-readable definitions of these vocabulary entries are provided in extraction_method.json.

Term	Definition
AI-assisted	Data extracted or interpreted using artificial intelligence tools, machine learning methods, large language models, computer vision systems, or AI-assisted workflows with subsequent human verification.
computed	Data obtained through analytical calculations, numerical processing, statistical analysis, inverse analysis, or other computational procedures applied to available information.
digitized	Data extracted through manual or automated digitization of plots, graphs, images, or scanned documents.
direct	Data transcribed directly from tables, appendices, supplementary materials, spreadsheets, or explicitly reported numerical values in the original source.
unknown	Extraction method not reported or not identifiable from the available documentation.

9.12.2 Data Types

The following table defines the controlled vocabulary for the `data_type` field used in the `data` tab. The machine-readable definitions of these vocabulary entries are provided in `data_types.json`.

Term	Definition
`2D_field`	Two-dimensional spatially distributed dataset such as displacement fields, strain fields, temperature fields, moisture fields, or DIC measurements defined over a planar domain.
`3D_field`	Three-dimensional spatially distributed dataset such as voxel data, tomography reconstructions, volumetric temperature fields, moisture fields, or other volumetric measurements.
`curve`	One-dimensional dataset describing the relationship between two or more variables, typically represented as ordered numerical pairs or time histories.
`image`	Static visual dataset such as photographs, microscopy images, crack maps, thermal images, DIC visualizations, or experimental setup illustrations.
`scalar`	Single numerical quantity or statistically aggregated value associated with a specimen, test, or material property.
`table`	Tabulated multidimensional dataset containing multiple quantities organized in rows and columns.
`video`	Time-dependent visual dataset documenting experiments, crack propagation, deformation processes, failure evolution, or other dynamic phenomena.
`other`	Data type that does not fall within the predefined categories.

9.12.3 Reported Quantities

The following table defines the controlled vocabulary for the `quantity_reported` field used in the `data` tab. The machine-readable definitions of these vocabulary entries are provided in `reported_quantities.json`.

Term	Definition
`absorption`	Quantity associated with fluid absorption or uptake within the material.
`bound_water`	Quantity associated with chemically or physically bound water within the material.
`CMOD`	Crack mouth opening displacement.
`CTOD`	Crack tip opening displacement.
`coefficient_of_thermal_expansion`	Coefficient describing thermal expansion per unit temperature change.
`concentration`	Quantity associated with concentration of chemical species or constituents.
`cumulative_heat`	Accumulated heat release generated during hydration or thermal processes.
`current`	Electrical current quantity.
`deflection`	Displacement associated with structural bending or deformation response.
`density`	Mass density of the material or constituent phase.
`depth`	Geometric or penetration depth quantity.

Term	Definition
diameter	Diameter dimension associated with geometric or material features.
diffusion_coefficient	Coefficient describing diffusive transport processes.
displacement	Translational displacement quantity.
distribution	Spatial or statistical distribution of a measured quantity.
energy	Energy-related quantity such as dissipated energy or fracture energy.
expansion	Expansion or volumetric growth quantity.
fraction	Fractional quantity or proportion.
frequency	Frequency associated with cyclic loading, vibration, or dynamic response.
heat_capacity	Heat capacity or specific heat quantity.
heat_flow	Heat transfer rate or heat flux quantity.
humidity	Relative humidity or moisture-related quantity.
length_change	Variation in specimen length due to deformation, shrinkage, creep, or expansion.
load	Applied force or loading quantity.
mass	Mass quantity associated with the specimen or constituent materials.
mass_loss	Reduction in mass due to degradation, drying, leaching, or chemical processes.
permeability	Permeability associated with fluid or gas transport through the material.
pH	Hydrogen ion concentration expressed as pH.
porosity	Volume fraction associated with pore space within the material.
potential	Electrical or electrochemical potential quantity.
pressure	Pressure quantity associated with fluids, gases, or applied loading.
resistivity	Electrical resistivity quantity.
rotation	Angular displacement or rotational response quantity.
slippage	Relative displacement associated with sliding or bond-slip behavior.
slump	Fresh-state slump or workability quantity.
sorptivity	Sorptivity coefficient associated with capillary absorption processes.
strain	Mechanical strain quantity.
stress	Mechanical stress quantity.
temperature	Temperature quantity.
temperature_rise	Increase in temperature relative to a reference state.
thermal_conductivity	Thermal conductivity quantity.
thermal_diffusivity	Thermal diffusivity quantity.
time	Time quantity associated with measurements or loading histories.
viscosity	Viscosity or rheological resistance quantity.
volume	Volume quantity associated with material or specimen geometry.

Term	Definition
yield_stress	Stress associated with onset of yielding or flow.
other	Reported quantity that does not fall within the predefined categories.

9.12.4 Quantity Reported Types

The following table defines the controlled vocabulary for the quantity_reported field used in the data tab. The machine-readable definitions of these vocabulary entries are provided in quantity_reported.json.

Term	Definition
derived	Quantity obtained through post-processing, analytical calculations, inverse analysis, homogenization, curve fitting, numerical interpretation, statistical processing, or other forms of data reduction applied to experimentally measured data.
raw	Quantity corresponding directly to experimentally measured or recorded data without substantial post-processing or interpretation.
unknown	Reporting type not specified or not identifiable from the available documentation.

Chapter 10

Database Search Application

Part III

Computational Database

under development

Part IV

Parameter identification Application

under development

Part V

Validation Application

under development

Part VI

benchmarking Application

under development